

REVIEW

Open Access



A review of recent advances in the application of machine learning algorithms for gas turbine combustion

Da Mo^{1,2}, Yuzhen Lin^{2,3}, Yixiong Liu⁴, Yuchen Wang², Ziyu Qin² and Xiao Han^{2,3*}

Abstract

With the rapid advancement of computing power and significant progress in machine learning (ML) algorithms, ML has shown immense potential across a wide range of fields, particularly in simulation, experimental analysis, and condition monitoring. In the realm of combustion science and engineering, ML techniques have driven substantial advancements, particularly in gas turbine engine combustion. The integration of intelligent algorithms in combustion modelling and analysis has led to more accurate predictions, enhanced performance, and the potential for more efficient and environmentally friendly combustion processes. This paper provides a comprehensive review of the current state of research on intelligent algorithms applied to combustion chemical reaction kinetics, combustion simulation, performance prediction, combustion state, and instability monitoring. The review highlights the progress made and offers valuable insights for improving the performance of gas turbine engines. Additionally, the paper discusses the challenges and prospects of applying intelligent algorithms in combustion research, including issues related to data quality, model interpretability, and computational complexity, while identifying avenues for future development and innovation.

Keywords Machine learning, Chemical reaction kinetics, Instability prediction, PINN

1 Introduction

In the current data-driven era, machine learning (ML) algorithms are continuously advancing technological development. These algorithms, which simulate intelligent behaviors observed in nature, such as learning, reasoning, and perception, are designed to solve complex problems. Examples of such algorithms include linear regression (LNR), logistic regression (LGR), decision tree

(DT), random forest (RF), neural network (NN), convolutional neural network (CNN), recurrent neural network (RNN), support vector machine (SVM), and generative adversarial network (GAN). Due to their versatility, these algorithms hold significant potential in a wide range of applications, such as prediction, classification, pattern recognition, and data generation [1, 2].

LNR and LGR serve as fundamental statistical modeling approaches due to their computational efficiency and model transparency. LNR establishes linear correlations between independent variables and continuous responses, making it particularly effective for quantitative predictions such as temperature forecasting. In contrast, LGR employs a sigmoid transformation to estimate class probabilities, demonstrating superior performance for classification tasks with limited datasets or high-dimensional sparse features.

*Correspondence:

Xiao Han

han_xiao@buaa.edu.cn

¹ AECC Harbin Dongan Engine Co., LTD., Harbin 150066, China

² Research Institute of Aero-Engine, Beihang University, Beijing 100191, China

³ National Key Laboratory of Science and Technology on Aero-Engine Aero-thermodynamics, Beihang University, Beijing 100191, China

⁴ AECC Shenyang Engine Research Institute, Shenyang 110015, China

Decision trees and RFs excel in handling nonlinear relationships and high-dimensional data, though they may face the risk of overfitting. Decision trees partition the feature space into regions using a series of decision rules. Each internal node represents a decision based on a feature, and each leaf node represents a prediction. RF is an ensemble method that constructs multiple DTs. Each tree is trained on a random subset of the data and a random subset of features. The final prediction is aggregated by majority voting (classification) or averaging (regression).

NNs consist of layers of interconnected neurons that process input data through weighted sums and activation functions. NNs, particularly deep learning (DL) models, have become a research focus due to their ability to handle complex data relationships, though they require vast amounts of data and computational resources. SVMs identify the optimal hyperplane that maximizes the margin between different classes. For non-linearly separable data, they employ kernel functions to map the data into higher-dimensional spaces, enabling effective classification and regression. This approach allows SVMs to process data indirectly via kernel functions, thereby avoiding the computational complexity of direct high-dimensional operations. It is especially advantageous for handling problems where data cannot be separated by a linear boundary.

NNs are computational models that simulate the structure and function of the human brain, capable of processing complex data relationships, particularly in fields such as image recognition, natural language processing, and speech recognition. NNs predict and classify by learning the intricate patterns and relationships between input data. Numerous variants of NNs exist, with CNNs achieving revolutionary advancements in image recognition [3]. The introduction of convolutional and pooling layers in CNNs effectively addresses the dimensionality curse,

spatial feature loss, and gradient issues encountered by artificial neural networks (ANNs) when processing high-dimensional data. These innovations enable CNNs to excel in image classification and computer vision tasks, especially in analysing complex flame structures. RNNs process sequential data by updating hidden states based on current inputs and previous hidden states. They are prone to vanishing or exploding gradients, which are mitigated by variants, such as long short-term memory (LSTM) networks. They perform exceptionally well in processing natural language and time-series predictions. GANs consist of a generator that produces synthetic data and a discriminator that distinguishes between real and fake data. They are trained adversarially until the generator produces data indistinguishable from real data. This process enables GANs to generate high-quality, realistic data samples, particularly in image synthesis.

The development of ML algorithms marks a significant advancement in computational technology, as presented in Fig. 1. Machine learning techniques have evolved from fundamental algorithms (including linear regression [4] and neural networks [5]) to more advanced approaches like deep learning [6, 7] and reinforcement learning [8], with applications extending to domains such as chemical kinetics modeling via Arrhenius equations [9]. The field has advanced significantly with the rise of big data, GPU acceleration, and DL frameworks. The development of transformer architectures and foundation models has further advanced ML capabilities, enabling a broad range of applications. Transformers, with their powerful attention mechanisms, have revolutionised natural language processing and have also found applications in other domains such as computer vision and multimodal learning. Their ability to capture long-range dependencies and parallelize computations makes them highly effective, though they come with challenges like high computational complexity and memory requirements. On the

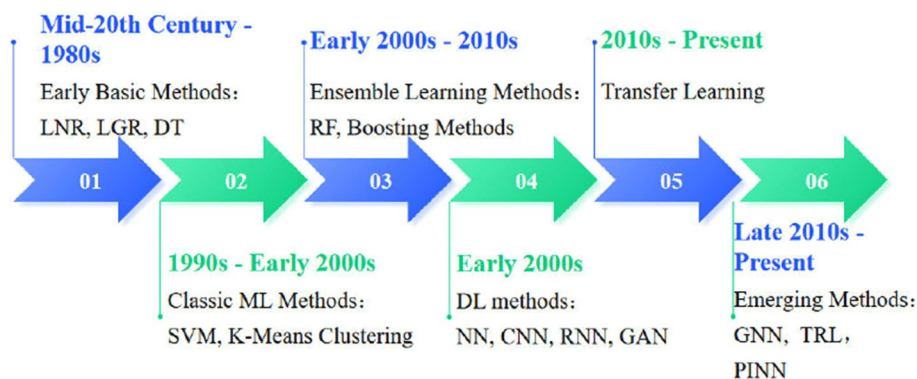


Fig. 1 ML algorithms development

other hand, GNNs are designed to handle graph-structured data, excelling in tasks like node classification, link prediction, and recommendation systems by leveraging the inherent relationships between nodes. Despite their strong modeling capabilities, GNNs face issues like over-smoothing and high computational costs for large-scale graphs. The future of ML looks to be defined by multi-modal models that integrate diverse data types for more complex problem-solving tasks.

The development of ML algorithms has brought revolutionary changes to various fields, showcasing immense potential in improving efficiency, reducing costs, and uncovering new knowledge. With the growth in computational power and the explosion of data, ML algorithms will continue to play a crucial role across many domains, driving technological progress. In the field of combustion, the application of ML plays a critical role in turbulence modelling, chemical reactions, and experimental research.

The following provides an overview of the development and application of these technologies. For turbulence modelling, ML and DL techniques have proven to be effective tools. Researchers have used these methods to predict the characteristics of turbulent flows, such as Reynolds stress models [10]. Spalart introduced turbulence models and simulation strategies and presented a new version of the detached eddy simulation (DES) that is resistant to grid resolution variations [11, 12].

Regarding chemical reaction modelling, ML technologies are used to simulate and predict complex chemical reactions [13–15]. Key areas of focus include combustion mechanism optimization [16, 17], chemical reaction kinetics modelling, and computational acceleration [18–21]. ANNs have been successfully utilized to simulate chemical reactions [22]. These networks have shown the capability to capture the nonlinear functions of temperature and concentration. Additionally, unknown reaction pathways have been autonomously discovered through the use of chemical reaction NNs [23]. Moreover, complex reaction processes have been revealed by employing molecular dynamics simulations [24].

Concerning numerical simulations, ML techniques are employed to improve the accuracy of combustion efficiency predictions and optimize combustion processes. Buchheit et al. [25] proposed a stabilized explicit variable-load solver that accelerates the solution of stiff chemical kinetics problems using ML. Chen et al. [26] applied ML to the closure of the filter density function in moderate or intense low-oxygen dilution (MILD) combustion.

As for experimental research, ML technologies are also used to assist with sensor data augmentation and combustion process control [27–30], as well as performance prediction, fault diagnosis, health management, and performance optimization [31–33]. ML further aids

in analyzing vast amounts of combustion test data to identify key parameters in the combustion process, enabling prediction, diagnosis, and management [34, 35]. To address high noise in industrial data and improve robustness under varying operating conditions, various techniques could be used, such as regularization methods like L1 and L2 regularization. Feature engineering and selection allow models to focus on the most relevant features, minimizing the influence of noisy or irrelevant data. Data augmentation techniques, such as adding noise, rotation, and scaling, enhance the model's ability to generalize by exposing it to diverse variations during training. Dropout and batch normalization in NNs further improve robustness by preventing overfitting and stabilizing the training process. Ensemble methods like RF reduce variance and improve generalization by averaging multiple models. Kernel functions in SVMs and attention mechanisms in RNNs help handle non-linearities and focus on relevant parts of the data, enhancing performance under varying conditions.

This paper aims to provide a comprehensive review of the application of intelligent algorithms in the field of gas turbine engine combustion. It explores how these algorithms assist researchers in better understanding and predicting combustion phenomena and addresses the challenges and future directions for their practical implementation.

2 Combustion modelling

ML with combustion models is crucial for enhancing the accuracy, adaptability, and optimization of complex gas turbine combustion processes. By learning from large datasets, they refine predictive accuracy, adapt to varying conditions, and reduce model uncertainties. This combination offers significant advantages over traditional models, enabling more efficient operation in practical applications.

2.1 Turbulent combustion models

ML in turbulent combustion models primarily improves prediction accuracy and computational efficiency, addressing the limitations of traditional models in handling complex nonlinear and high-dimensional data. Techniques such as DL and neural networks enable the optimization of reaction rate predictions, turbulence-combustion coupling, and low-dimensional data modelling. Recent studies have made significant progress in developing efficient turbulent combustion models, real-time simulations, and multi-scale modelling.

The integration of ML into the combustion model has proven effective in improving prediction accuracy and computational efficiency [36, 37]. The ANN model outperforms traditional wall models in terms of accuracy,

particularly under boundary layer conditions characteristic of turbulent combustion, as depicted in Fig. 2 [38].

CNN could efficiently extract the flame topological features and predict subgrid-scale wrinkling, outperforming traditional algebraic models [39]. Additionally, CNN exhibited excellent deconvolution performance, recovering lost small-scale information while maintaining low error rates across different test scenarios [40]. Furthermore, CNN and a hybrid downsampling skip multiscale model have demonstrated great potential in the super-resolution reconstruction of low-resolution turbulent flow field data [41]. The generalized $k-\omega$ turbulence model was tuned for gas turbine combustor applications

using adjoint optimization and NNs, as shown in Fig. 3 [42]. By validating against large eddy simulation (LES) and experimental data, it showed consistency in velocity field variance and turbulence energy terms.

The GAN method in modelling sub-filter terms in turbulence was proven to achieve good results in prior and posterior tests, including decaying turbulence and turbulent mixing [43], as shown in Fig. 4. The physical information continuity loss term was also important.

ML approaches are utilized in RANS and LES modeling to represent unclosed terms, model discrepancies, and subgrid-scale phenomena [44]. Shin et al. [45] proposed a novel probabilistic data-driven approach

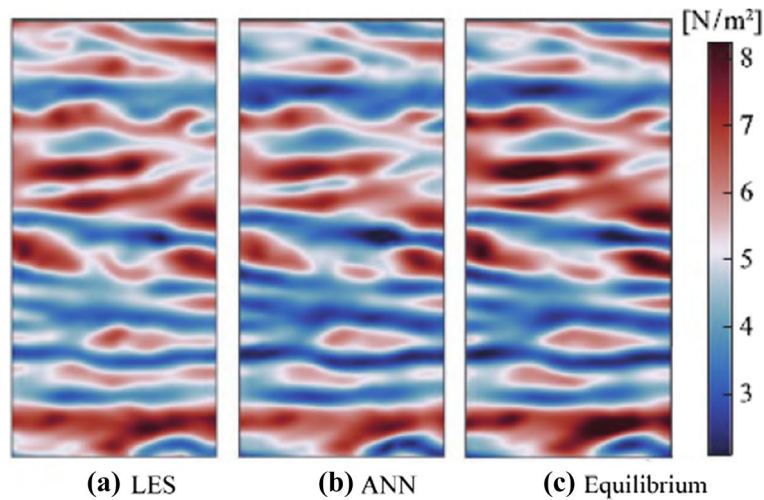


Fig. 2 Results comparisons

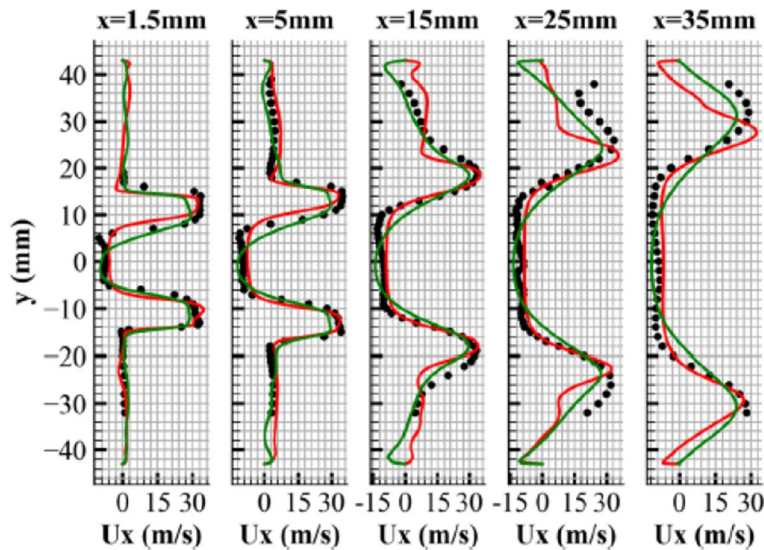


Fig. 3 Axial velocity distribution for reacting flow compared to experiments

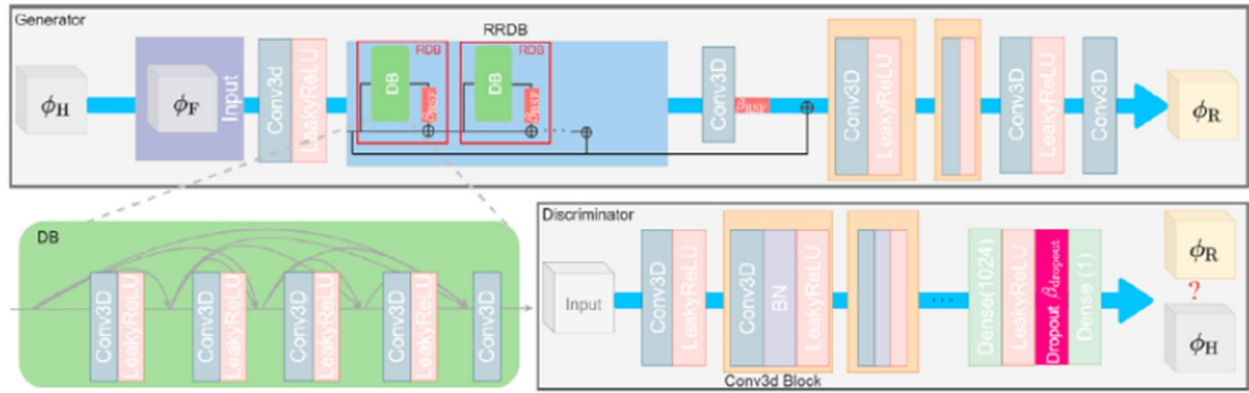


Fig. 4 Diagram of the GAN architecture

for simulating filtered reaction rates in LES of turbulent premixed combustion. This framework integrates conditional generative adversarial networks (cGANs) and Gaussian mixture models to account for statistical fluctuations in LES caused by unresolved subgrid structures. The study demonstrated that probabilistic models are significant in capturing the stochastic nature of turbulent combustion and showed that the proposed model outperformed traditional deterministic models.

Additionally, Shin et al. [46] introduced a data-driven subgrid-scale model for turbulent premixed combustion in LES. The study employed an architecture combining deep residual networks (ResNet) with mixture density networks (MDN) to estimate subgrid flame surface density using only local variables. The DNS data provided by Lapeyre et al. [47] were used for training, and the effectiveness and accuracy were validated through a priori tests and posterior assessments.

Shateri et al. [48] explored the integration of ML techniques with LES to predict species mass fractions and flame characteristics in partially premixed turbulent jet flames. To enhance real-time prediction capabilities, three ML models were trained: NN, LNR, and decision tree regression (DTR). The NN model demonstrated high precision in predicting species mass fractions and flame patterns and achieved a computational efficiency improvement of approximately 17.25 times.

Seltz et al. [49] proposed a unified modelling framework that uses CNN to provide closure for all unresolved terms in LES, as shown in Fig. 5. The study demonstrated that the trained networks could produce quantitatively good predictions of all unresolved terms in prior studies, without requiring the solution of any additional transport equations.

As depicted in Fig. 6, Wan et al. [50] introduced a novel CNN-based reduction strategy for DNS of turbulent non-premixed flames with cooled walls, utilizing syngas

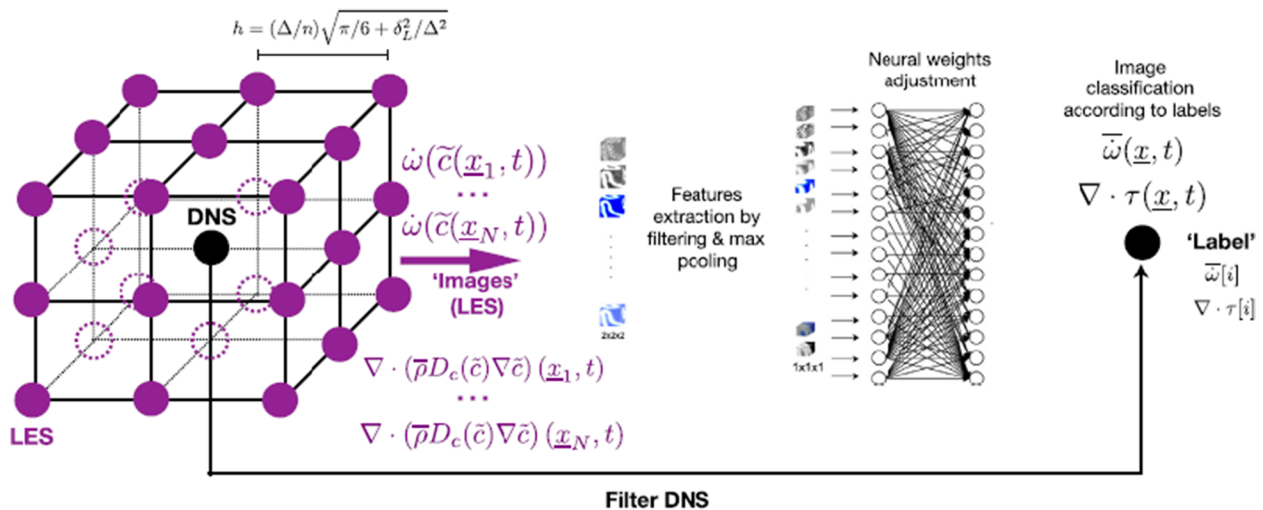


Fig. 5 CNN training from DNS, the sketch of the database construction

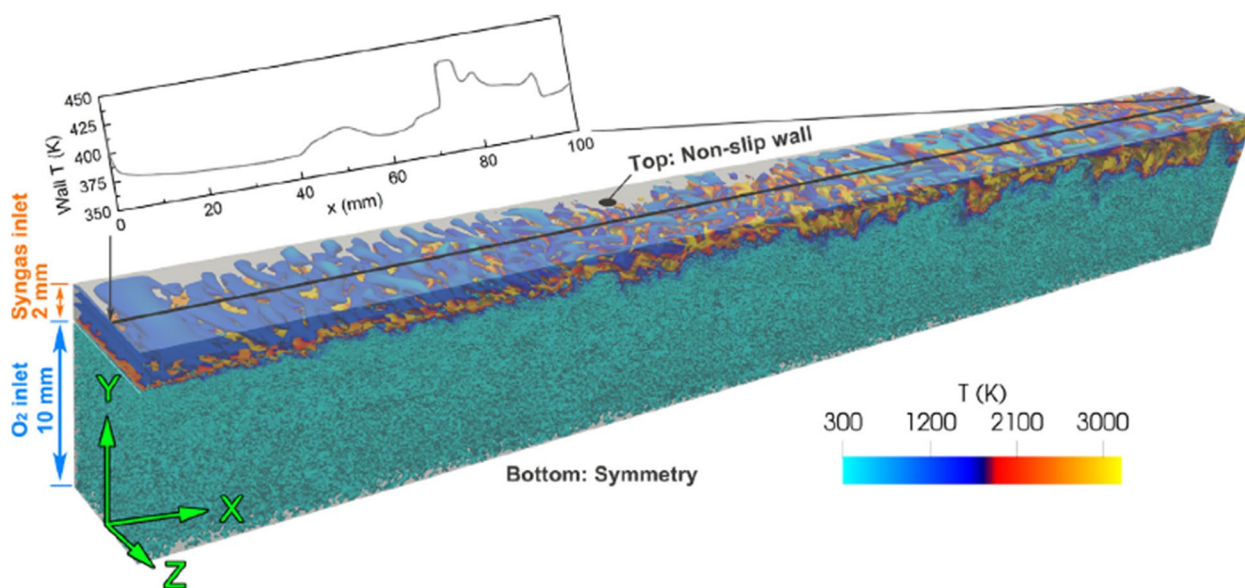


Fig. 6 Instantaneous iso-surface of Q criterion predicted by CNN-DNS [50]

in pure oxygen. By processing 2D images of species mass fractions and temperature, the CNN effectively predicted chemical source terms, outperforming traditional Arrhenius-based mechanisms in accuracy for radical species, as presented in Fig. 7.

A PCA-ANN combustion model was developed to map thermochemical states using five key variables, trained on data from homogeneous reactors, non-premixed flame slices, and 2D-lifted flames [51]. It can accurately predict the major species of turbulent jet flames based on DNS, as shown in Fig. 8, but improvements are still needed in predicting minor species and reaction rates. This requires further enhancement of the training data to address these issues.

The study of flame propagation speed is crucial for understanding and predicting combustion processes, as it involves the chemical reaction properties of the fuel,

heat release characteristics, and the stability of the flame under various conditions, which in turn helps ensure safety during combustion. With the growing demand for clean energy, studying the flame propagation characteristics of new fuels, such as hydrogen, has become increasingly important. Adding hydrogen to propane combustion can enhance flame stability, widen the lean flammability limits, and reduce pollutant emissions. For the laminar burning velocity of propane/hydrogen/air mixtures, ANN showed superior accuracy compared to SVM and significantly faster computation times compared to traditional methods [52]. Moreover, using laminar flame speed calculations to train ML models enabled faster combustion rate predictions within a virtual engine framework [53], as shown in Fig. 9.

Compared to multiple models of linear regression (MLR), SVR, RF, and ANN, the RF model provided the

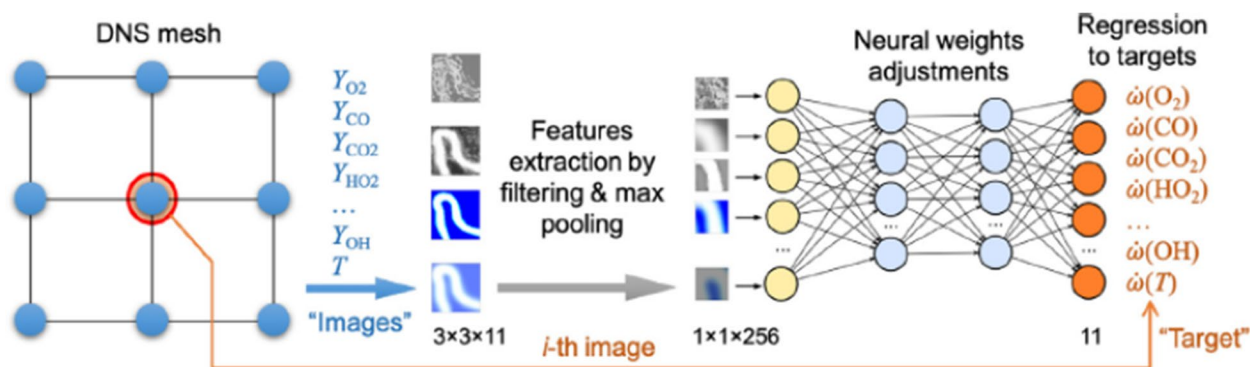


Fig. 7 CNN training from a 2D database [50]

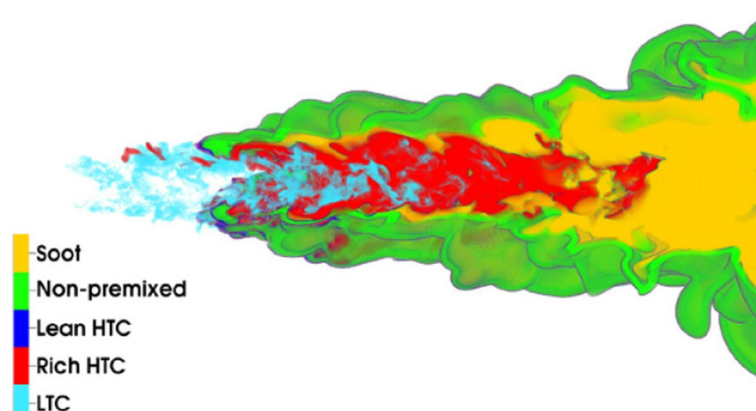


Fig. 8 Flame coloured by combustion modes [51]

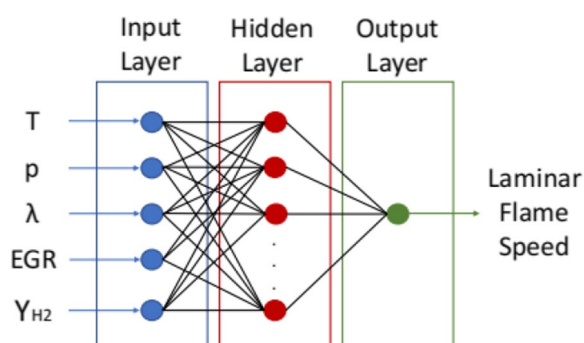


Fig. 9 ANN architecture [53]

highest accuracy for predicting turbulent flame speeds in hydrogen-natural gas spark ignition engines [54] (Fig. 10).

2.2 Combustion kinetic model

In the context of chemical reaction kinetics modeling [55–58], ML techniques have been employed to construct and simplify complex chemical reaction mechanisms, thereby reducing computational costs and enhancing computational efficiency. For instance, optimizing chemical reaction mechanisms using neural networks can significantly lower computational expenses while maintaining high prediction accuracy. Using optimization algorithms, the accuracy of ANNs can reach the level of medium-to-fine resolution tabulated methods [59]. The ANN-based approach could accelerate global sensitivity analysis of combustion kinetics models and require only a fraction of the original samples [60].

Reaction rate constants are essential parameters in chemical kinetics models, and their accurate prediction is crucial for combustion simulations. Lu et al. [61] utilized machine learning to predict the rate constants of hydroxyl radical reactions with alkanes. By constructing

machine learning models and combining extensive experimental data with theoretical calculations, they achieved efficient prediction of reaction rate constants. Li et al. [56] also employed machine learning to predict rate constants for various reactions in combustion kinetic models, further demonstrating the potential of machine learning in rate constant prediction, which can significantly improve prediction accuracy and reduce computational time. Additionally, KiNet [62], a deep neural network representation of chemical kinetics, models chemical reaction kinetics by automatically learning the complex relationships between reaction rate constants and reaction pathways, offering a new tool for combustion chemical kinetics modeling and simulation (Figs. 11 and 12).

Optimizing combustion kinetic mechanisms is a key step in improving combustion simulation accuracy. A study [57] proposed a machine learning-based optimization method for hydrogen combustion kinetic mechanisms. By analyzing and mining extensive experimental data using machine learning algorithms, the kinetic parameters were optimized, resulting in a more accurate representation of hydrogen combustion processes. Liu et al. [63] developed Reaction Mechanism Generator v3.0, which has made significant progress in automatic mechanism generation. This tool uses machine learning to intelligently select and optimize reaction pathways and kinetic parameters, enhancing the accuracy and reliability of generated mechanisms and providing strong support for the development of kinetic mechanisms in complex combustion systems.

Deep learning, a significant branch of machine learning, has also been widely applied in combustion chemical kinetics. A study [64] explored the application of deep learning in analyzing and reducing dynamic latencies in combustion chemical kinetics. By analyzing

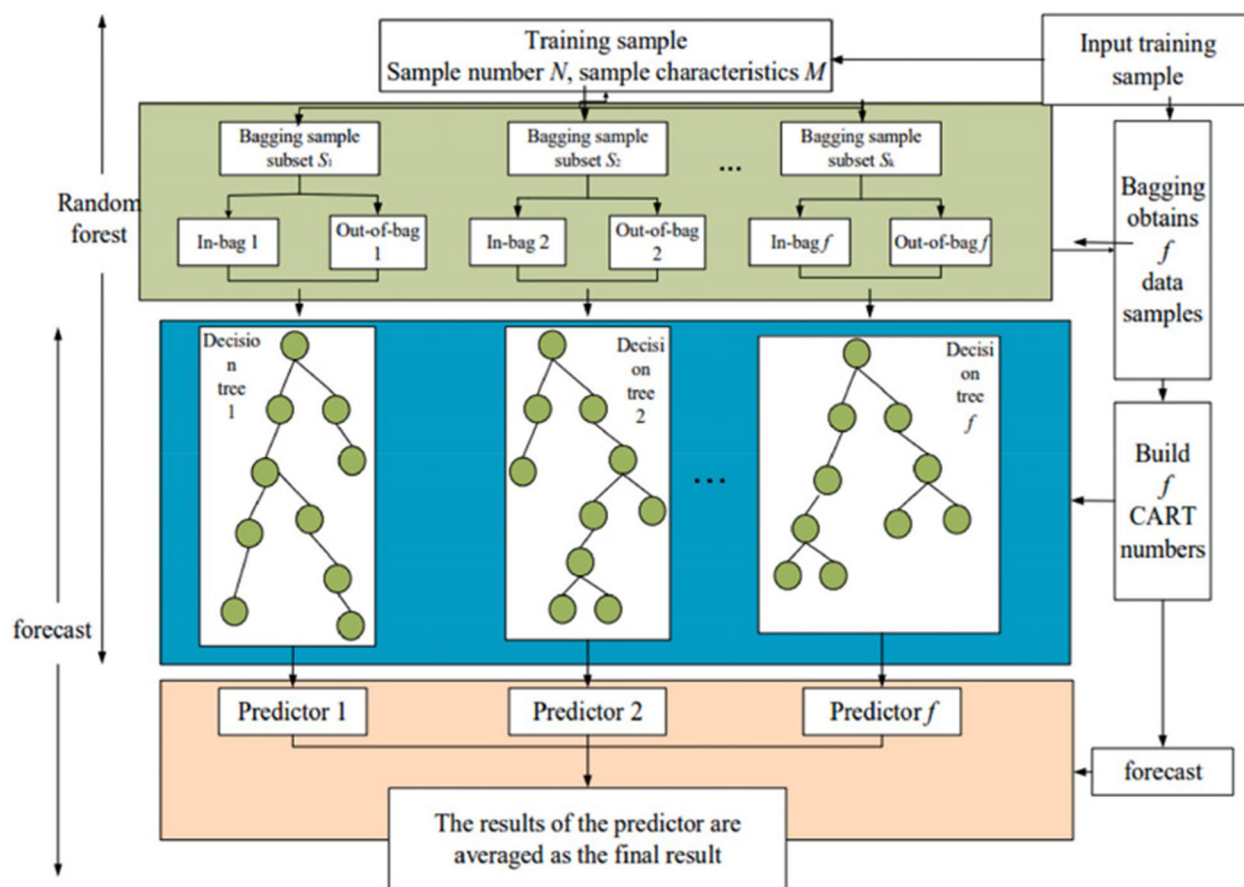


Fig. 10 Random forest training [54]

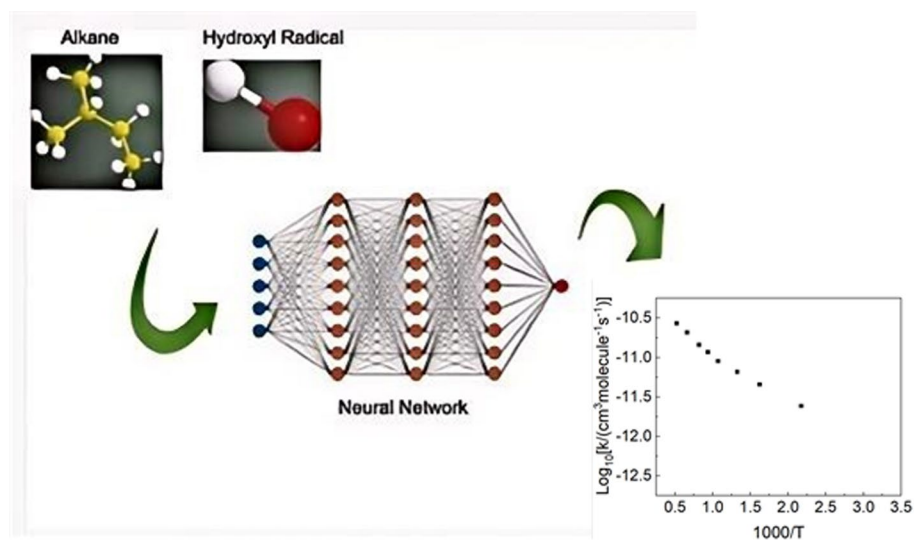


Fig. 11 Schematic of the rate constants prediction [61]

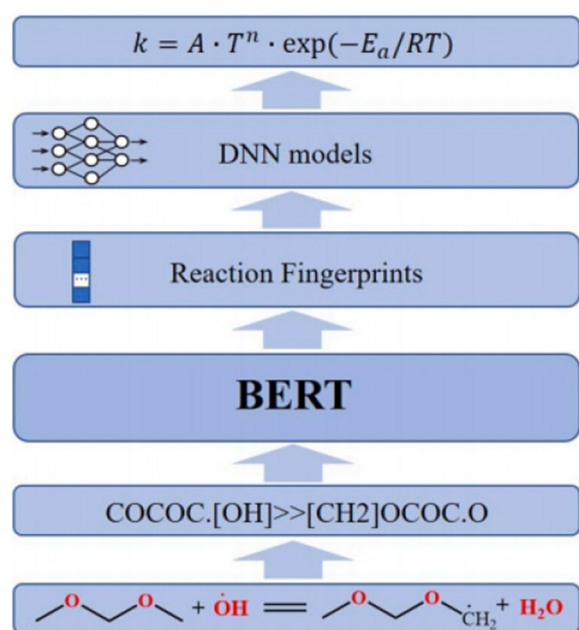


Fig. 12 Flowchart for using reaction fingerprints to predict rate constants [56]

time delays and reaction pathways using deep learning models, hidden dynamic characteristics in combustion processes were revealed, providing new insights into understanding complex combustion chemical kinetics behaviours. Han et al. [65] proposed an improved deep-learning model to enhance the robustness of chemical kinetics models. By optimizing the deep learning network structure and training algorithms, the model maintained high prediction accuracy under different operating conditions and data noise, enhancing the reliability of chemical kinetics models in practical applications (Fig. 13).

Combustion chemical kinetics models often contain a large number of reactions and species, resulting in high computational costs. Therefore, model simplification and dimensionality reduction have become important research directions in combustion kinetics. Haghshenas [66] introduced the learned intelligent tabulation (LIT) method to accelerate chemical kinetics computations. By using machine learning to reduce the complexity of kinetic models, high-dimensional chemical kinetics problems were transformed into low-dimensional tabulated data, significantly improving computational efficiency while maintaining model accuracy. Pulga et al. [67] employed machine learning to improve the accuracy of reduced chemical kinetics mechanisms in laminar flame simulations. Their study optimized the simplified mechanisms using machine learning, resulting in higher accuracy in simulating laminar flame

propagation speeds and temperature distributions, providing a new approach for efficient combustion simulations (Fig. 14).

Machine learning techniques can also be used to integrate kinetic mechanisms from different sources, enhancing model applicability and accuracy. For example, by analyzing multiple experimental datasets and theoretical calculations using machine learning algorithms, more comprehensive and accurate kinetic models can be generated. This approach not only reduces the influence of human factors on models but also better adapts to the kinetic requirements under different combustion conditions. The application of artificial intelligence in the modeling of chemical reaction kinetics [68] has also been explored, highlighting the potential of artificial intelligence (AI) techniques to enhance the development and integration of kinetic models.

2.3 Conditional scalar dissipation rate modeling

Conditional Scalar dissipation rate (CSDR) is an important parameter for describing mixing and reaction processes in turbulent flow fields, particularly in the study of combustion and pollutant formation. Traditional CSDR modelling methods typically rely on empirical formulas or similar approximations. With the ML method, researchers can leverage large datasets from DNS or experimental data to train models, capturing complex nonlinear relationships and identifying the connections between CSDR and other flow field variables. ANN trained on carrier-phase DNS data could model CSDRs in turbulent spray flames, and a few droplet-related parameters were sufficient for accurate predictions [69, 70], as shown in Fig. 15.

For modelling the reactive flows in LES, the DL model outperformed the analytical β - β PDF model and linear regression models in accuracy, while being as accurate as the RF model but with fewer trainable parameters and faster inference speeds [71]. CNN could effectively extract topological features of the flame and predict sub-grid-scale wrinkling as well as reaction rates in turbulent premixed flames [39, 72]. ANNs are capable of directly simulating an important unresolved term in the unresolved stress tensor [73]. Meanwhile, DNN can predict the dissipation rate of the progress variable in turbulent premixed flames [74].

Table 1 highlights the diverse approaches to combustion prediction, where intelligent algorithms like ANN, CNN, and NNs significantly improve accuracy. However, the use of high-quality data and the computational intensity required for large datasets and complex simulations remain critical factors for achieving robust results. These models demonstrate the evolving trend of integrating advanced ML techniques with classical combustion and

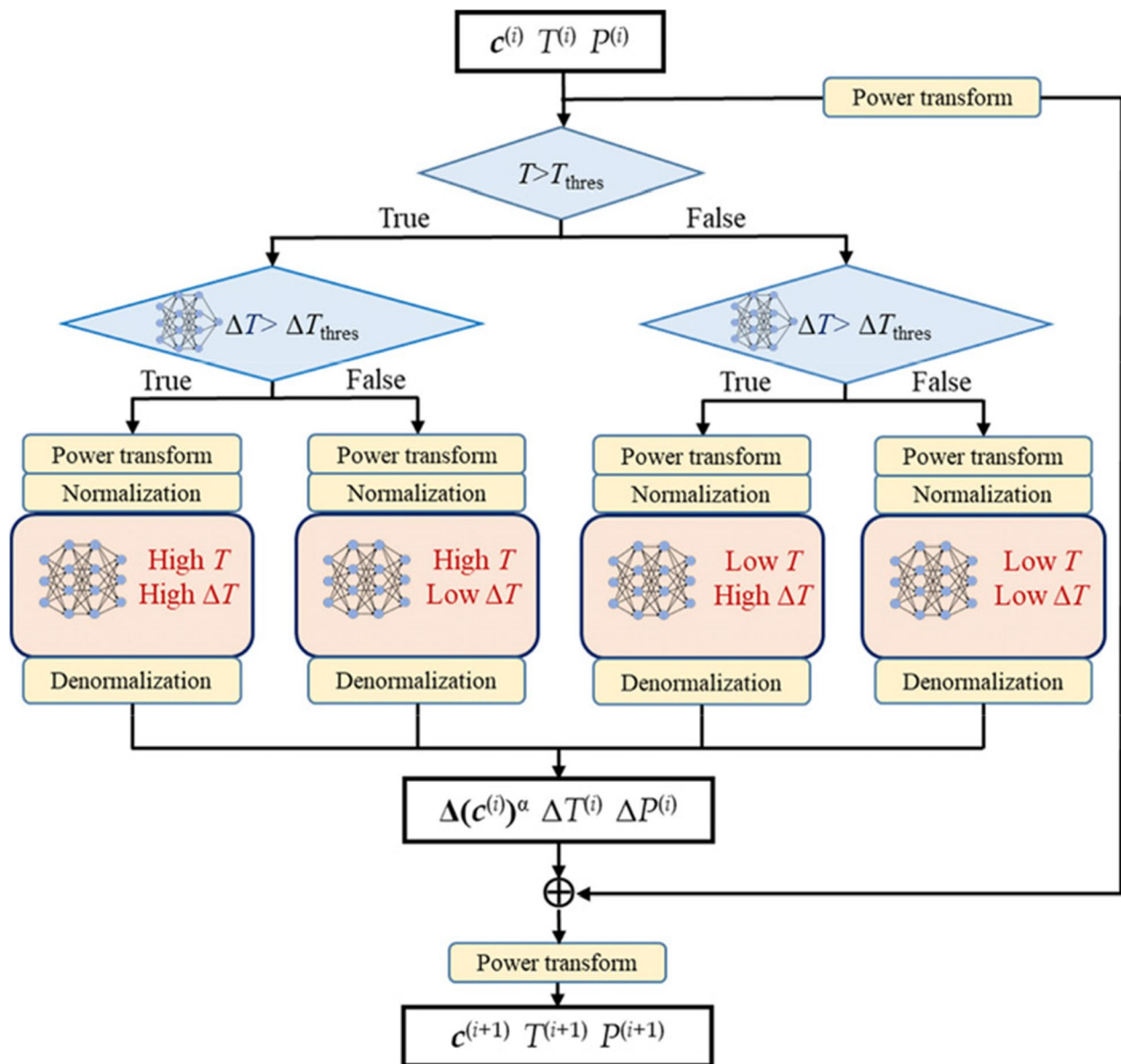


Fig. 13 Schematic of the model algorithm [65]

turbulence models to enhance predictive capabilities in combustion research.

3 Numerical simulation

Numerical simulation analysis plays a crucial role in various engineering fields, especially in applications involving combustion. They help to predict flame propagation, combustion states, pollutant emissions, and extinguishment processes without the need for physical experiments. With the rapid advancements in computational power and ML technologies, intelligent algorithms provide innovative approaches and tools for addressing

complex engineering challenges. These algorithms are particularly effective in tasks such as optimization, predictive modelling, and real-time control, further enhancing the accuracy, efficiency, and capabilities of traditional numerical simulations.

3.1 Combustion state identification

Combustion state identification refers to the process of identifying and characterizing the various phases of combustion within a combustion system. Accurate identification of the combustion state is crucial for optimizing combustion efficiency and reducing pollutant emissions.

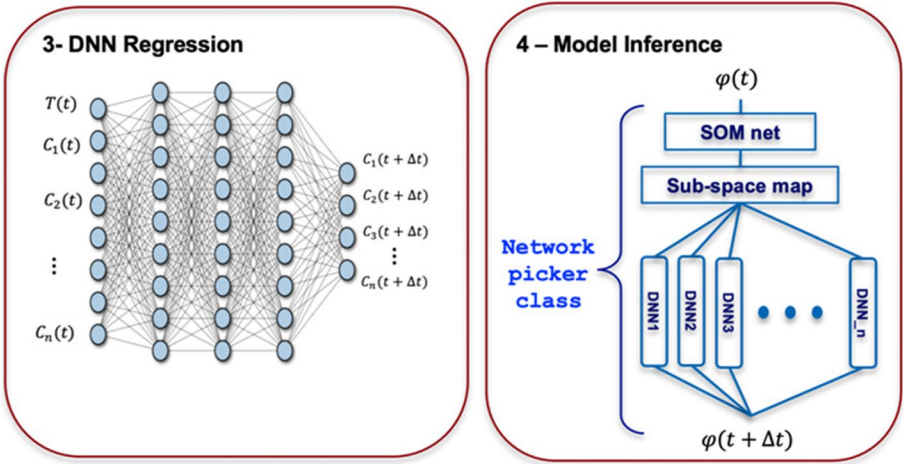


Fig. 14 The overview of the LIT method [66]

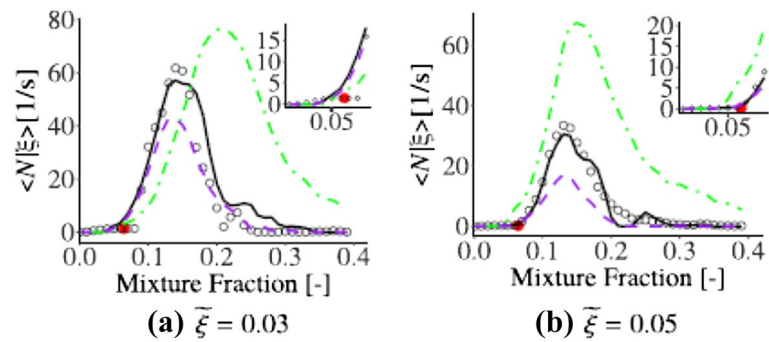


Fig. 15 Instantaneous CSDRs of fuel-lean LES cell [69, 70]

Table 1 The application of intelligent algorithms in the theoretical model

Author	Fuel	Data acquisition methods	Data volume	Method	Prediction accuracy	Theoretical model
Yu et al. [36]	Hydrogen	FlameMaster database	1.02 million +	ANN	Error reduction over 50% on average	Combustion model
Zheng et al. [38]	–	DNS	3.6 million	NNs	$R > 0.97$, RMSE = 0.501	Turbulent combustion model
Klavaris et al. [42]	–	ANSYS FLUENT	–	NNs	Accuracy improved by 70%	Turbulence model adjustment
Ihme et al. [59]	Methane and hydrogen	LES	synaptic weights 50000	ANN	Consistent with experimental results	Chemical characterization
Seltz et al. [49]	Methane	DNS	10800	CNN	Maximum error of 1%	Unified modeling framework
Wan et al. [50]	Synthesis gas and pure oxygen	DNS	–	CNN	–	Theoretical model
Yao et al. [69, 70]	Kerosene	CP-DNS	–	ANN	Error less than 15%	Turbulence model adjustment

By combining wall pressure data and Schlieren imaging, the DL model can reconstruct flame development, enhancing the identification of combustion modes across diverse chamber configurations [75]. Domain-adaptive NNs were used for dual-mode combustors, as presented in Fig. 16, and demonstrated improved

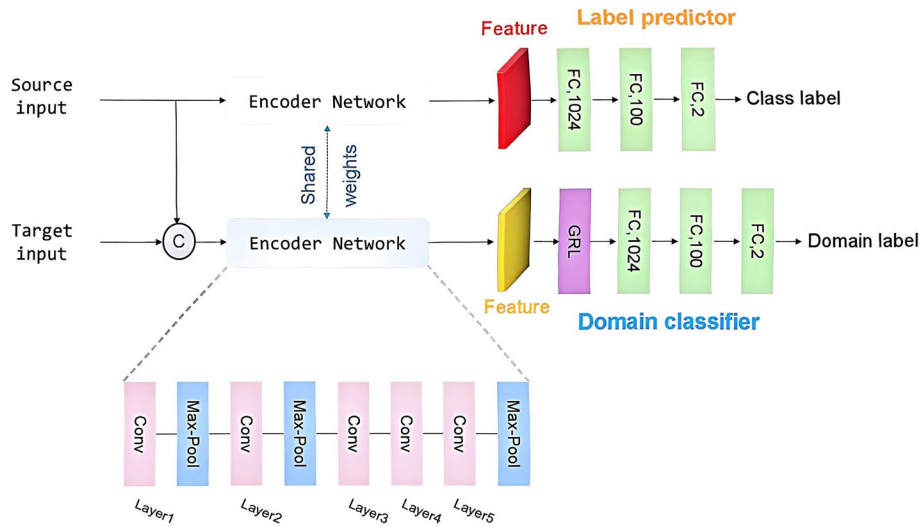


Fig. 16 Design framework

cross-domain recognition and generalization performance [76].

3.2 Pollutant emission prediction

Accurate forecasting of pollutants allows for the development of effective emissions control strategies, facilitating the optimization of combustion processes to reduce harmful emissions [77, 78]. Integrating ML and simulation techniques in pollutant prediction enhances our

understanding of complex combustion phenomena, ultimately aiding in the transition towards cleaner and more sustainable energy systems [79, 80].

RF classifier could be applied for dynamic submodel assignment in turbulent combustion simulations. It effectively assigned combustion models, rate chemistry (FRC), flame slices/progress variable (FPV) model, and inert mixing (IM), enhancing predictions of temperature and CO mass fraction [81] (Fig. 17).

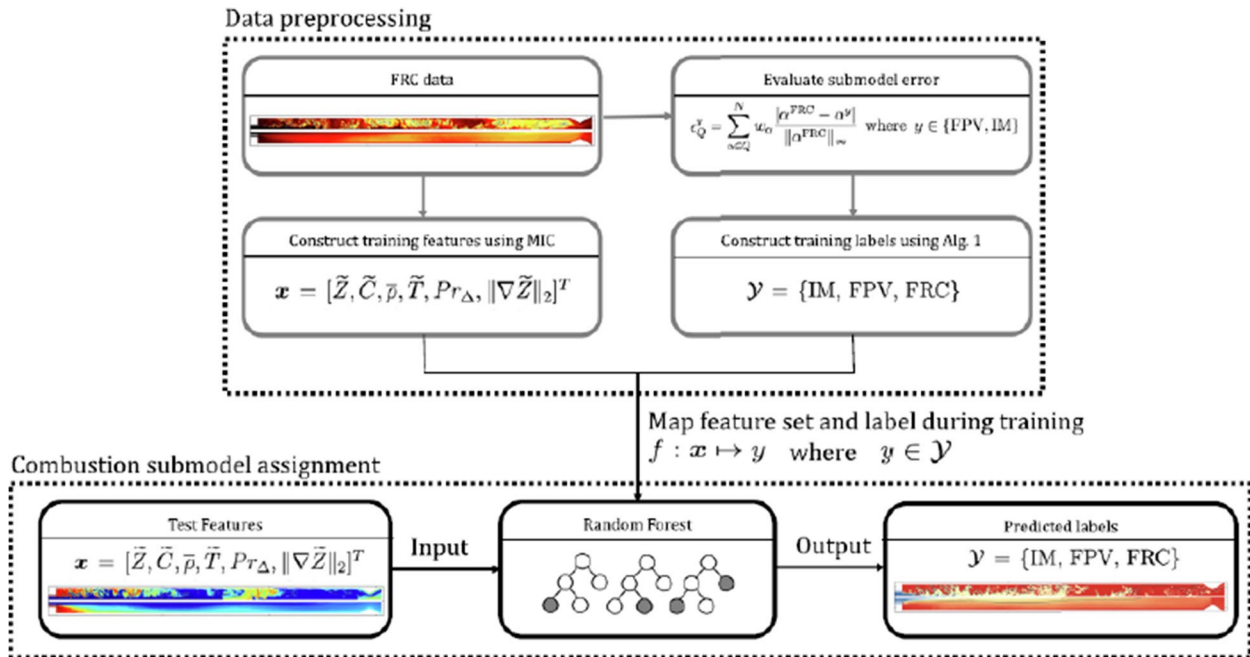


Fig. 17 Application of RF classifier for combustion submodel assignment [81]

3.3 Extinction prediction

Accurate extinction prediction is essential for combustor design, ensuring safety in industrial applications. Unlike traditional methods, ML models can handle high-dimensional data, adapt to varying conditions, and provide real-time predictions, making them highly suitable for complex combustion systems [82–84]. Moreover, they can enable early detection of lean blowout (LBO) [85]. For instance, an SVM model has been shown to identify the critical flame location and predict LBO 20 ms in advance, demonstrating its potential for online flame health monitoring and engine management [85], as shown in Figs. 18 and 19.

3.4 New emerging methods

Transformer, GNN, and physics-informed neural network (PINN) are advanced machine learning techniques that have revolutionized various domains by addressing

complex data structures and incorporating domain-specific knowledge.

Transformer methods have demonstrated remarkable potential in optimizing the performance of aviation engines and gas turbine combustion chambers. Specifically, a dual-stream network that integrates CNNs with Transformers has proven highly effective in classifying the combustion state of sintering flames. By seamlessly integrating these two streams, the model combines granular local insights with comprehensive contextual information, thereby enhancing its ability to accurately classify combustion states. The precision of this classification is further refined through the optimization of attention weights within the Transformer, as well as the implementation of multi-scale feature fusion techniques [86].

Additionally, Transformer-based deep neural networks have been employed to predict combustion reaction source terms, which are crucial for accurate

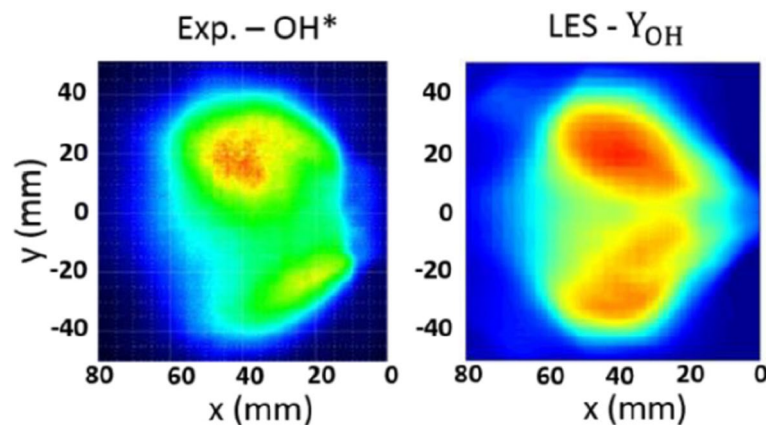


Fig. 18 Comparison of OH* chemiluminescence (experimental) and Y_{OH} (LES) [85]

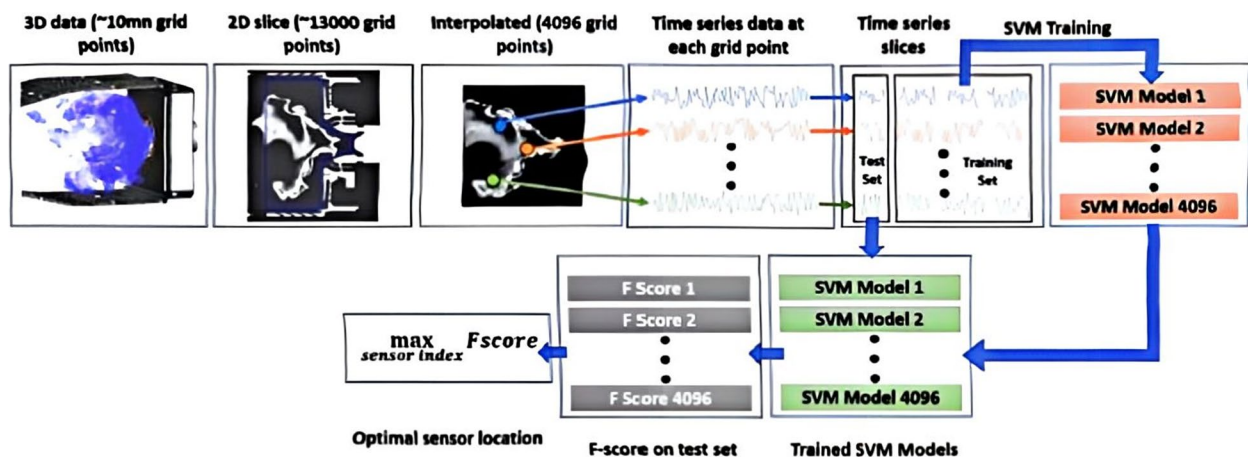


Fig. 19 Machine learning architecture [85]

combustion modeling [87]. These networks utilize a combination of encoder-decoder layers and self-attention mechanisms to handle complex, high-dimensional data. This architecture enables the model to not only process intricate patterns within the data but also provide highly accurate predictions.

GNNs could predict the ignition quality of fuels. The researchers trained the GNN using a diverse dataset of hydrocarbon compounds, optimizing the model to minimize prediction errors. The precision was enhanced through the use of advanced graph convolutional layers and attention mechanisms, which allowed the model to focus on critical substructures within the molecules [88]. The GREAD model is a novel application of GNNs designed to simulate reaction-diffusion processes, particularly relevant to combustion dynamics. This model integrates reaction-diffusion layers into the GNN framework, allowing it to capture both the spatial and temporal dynamics of combustion processes. This approach offers a more efficient and accurate alternative for modeling combustion dynamics, with potential applications in optimizing combustion processes and predicting flame behaviour [89].

PINNs have rapidly emerged as a transformative tool in the modeling of combustion processes, particularly for solving complex, multiphysics problems governed by partial differential equations (PDEs) [90]. Their applications span a wide range of combustion-related fields, including turbulent combustion, reactor design, and chemical kinetics. PINNs have demonstrated remarkable capabilities in accurately predicting key combustion parameters such as temperature distributions, species concentrations, and emission levels [91–94]. By integrating physical laws such as conservation of mass and energy into the model, PINNs provide a way to solve governing equations, like the Navier–Stokes equations and species conservation equations, with both data and physics constraints. These capabilities make PINNs well-suited for modelling the multiphysics phenomena inherent in combustion processes.

PINNs [95–97] offer several advantages in combustion modelling. Their ability to efficiently integrate sparse experimental or simulation data with physical laws leads to a more data-efficient and computationally cheaper approach compared to traditional methods. PINNs also ensure physically consistent predictions by enforcing governing equations during training. However, there are challenges to their widespread adoption. Training PINNs can be computationally expensive, particularly for large-scale combustion problems, and obtaining high-quality experimental data for training can be difficult. Furthermore, PINNs may struggle with generalization across diverse combustion conditions,

requiring careful training to handle a wide range of scenarios effectively.

Hybrid models that combine PINNs with traditional simulation techniques like LES could enhance predictive accuracy, especially for turbulent combustion regimes. Adaptive PINNs, capable of refining discretization based on problem complexity, may lead to improved performance in diverse scenarios. Additionally, PINNs could be adapted for real-time combustion monitoring and control, offering faster and more accurate predictions in dynamic systems. As research progresses, the ability of PINNs to handle multi-scale and multi-physics combustion phenomena would further expand their applicability, making them a valuable tool in the design and optimization of more efficient and sustainable combustion systems.

PINN for solving combustion reaction kinetics integrates physical principles to regularize the ANN, as demonstrated in Fig. 20. It can also infer intermediate species and temperature, and significantly reduce data dependence, achieving up to 14.6 times speedup in simulations and improving accuracy in reacting flow simulations [98].

Raissi [99, 100] introduces PINNs for solving supervised learning tasks while respecting physical laws. It presents two algorithms: continuous-time models for data-efficient spatiotemporal approximations and discrete-time models using high-accuracy Runge–Kutta schemes. The framework's effectiveness is demonstrated in classical problems in fluids, quantum mechanics, reaction–diffusion systems, and nonlinear wave propagation.

Table 2 summarizes the integration of ML with simulations for predicting combustion performance, featuring methods such as ANN, SVM, and CRK-PINN. While these algorithms provide significant advantages in terms of precision and efficiency, they rely on data quantity and quality, as well as computational resources.

4 Combustion experiment

The application of ML in combustion experiments has become increasingly necessary and important due to its ability to address the complexities and dynamic nature of combustion systems. As combustion processes are inherently nonlinear, multi-dimensional, and often influenced by numerous interacting variables, traditional modelling and experimental methods may struggle to capture the full range of behaviors. ML provides powerful tools to enhance the accuracy and reliability of combustion analysis, such as flow field reconstruction, flame reconstruction, anomaly detection, and combustion stability prediction.

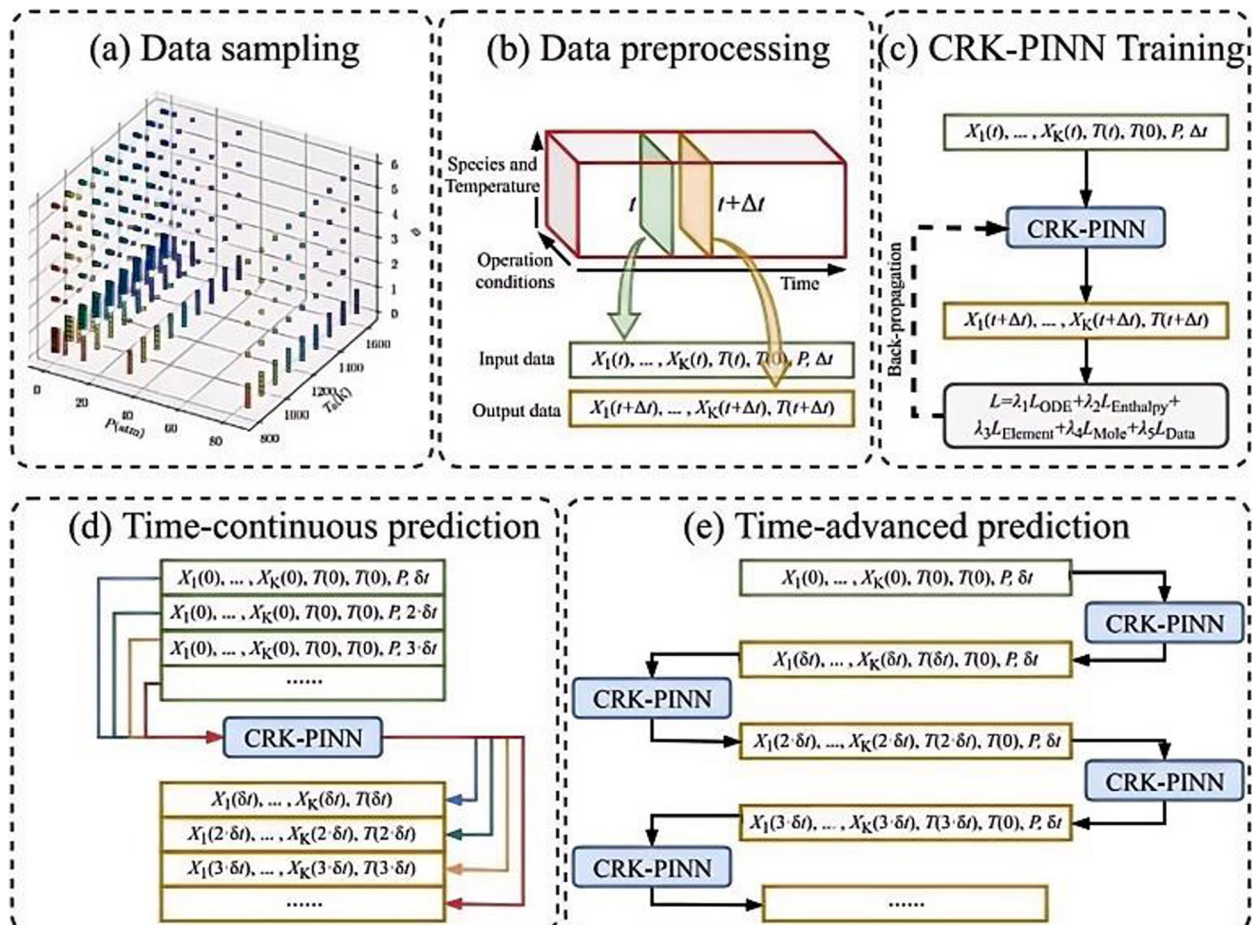


Fig. 20 The training process of CRK-PINN [98]

Table 2 Numerical simulation analysis situation

Author	Fuel	Method	Data volume	Prediction accuracy	Usage
Lu and Metghalchi [52]	Propane and hydrogen	Power law correlation, ANN, SVM	—	Approaching 100%	Laminar burning velocity
Molina et al. [53]	Methane-hydrogen	ANN	10,000	Error less than 0.0056	Burning rate
Song et al. [76]	Hydrogen	NN	1000	Over 90%	Combustion mode identification
Chung et al. [81]	GOX/GCH ₄	RFM	10000	—	Temperature and CO prediction
Hasti and Navarkar [85]	Kerosene	SVM	1300	F-score 0.94	Prediction of lean fuel shutdown
Zhang et al. [98]	Hydrogen	CRK-PINN	—	$R^2 > 0.99$	CRK

4.1 Flow field reconstruction

Flow field reconstruction offers critical insights into the fluid dynamics within combustion chambers. Key parameters such as velocity, pressure, and turbulence intensity significantly impact flame stability, heat transfer, and pollutant emissions.

DL has been applied to particle image velocimetry (PIV) flow field image restoration, demonstrating a high

confidence level in identifying abnormalities and adaptively repairing them while filtering noise and preserving the original data [101], as presented in Fig. 21.

NNs have been applied for particle tracking and edge analysis by early researchers such as Grant and Pan [102], while Lagemann et al. [103] used DL to achieve functionalities similar to traditional cross-correlation calculations. Zhu et al. [104] applied NNs to 3D particle

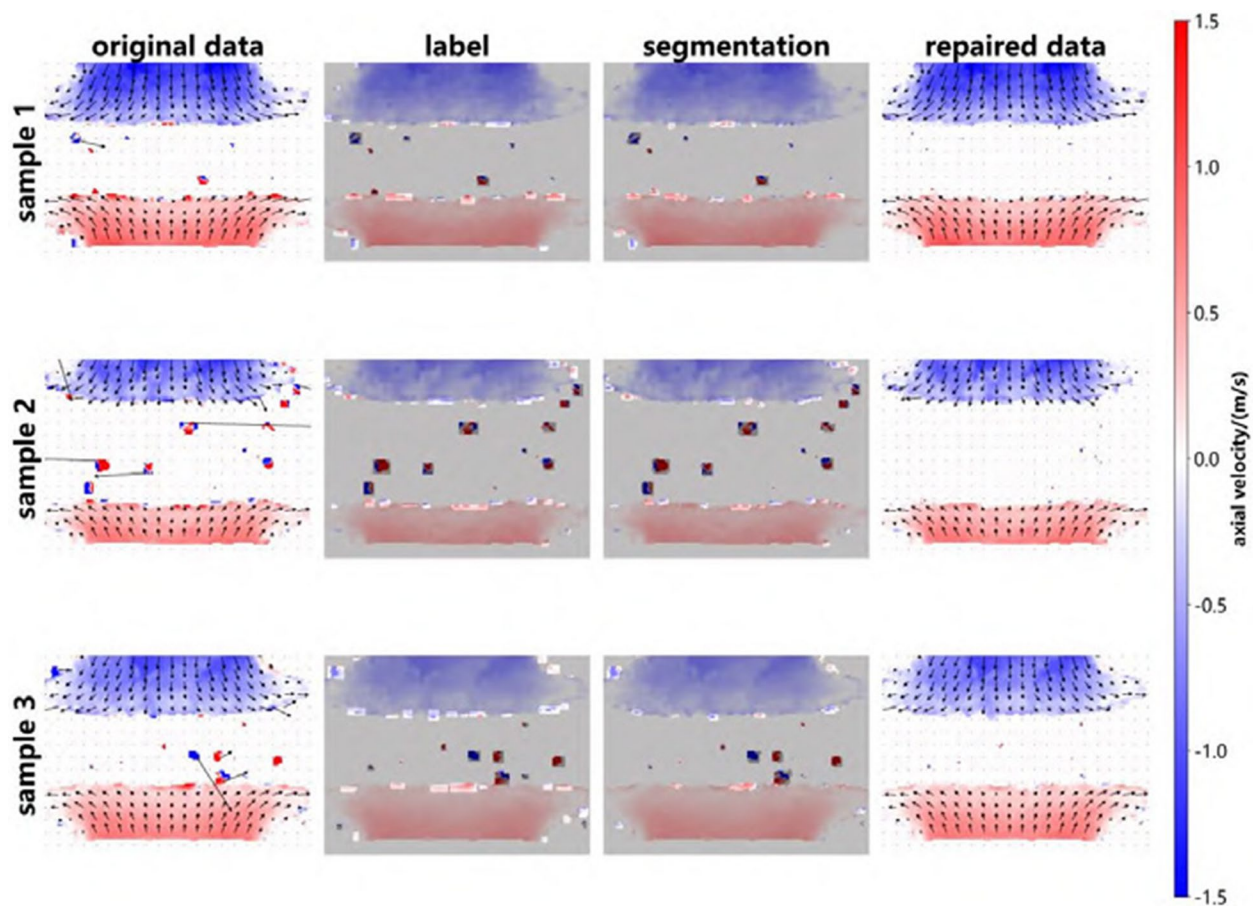


Fig. 21 Inpainting effect with different methods [101]

reconstruction in tomographic PIV, and Gao et al. [105] established optical distortion relationships in PIV measurements and constructed a dataset to eliminate optical errors using CNNs. At the University of Michigan [106], an NN method was employed to optimize the extraction of PIV data based on planar laser-induced fluorescence (PLIF) results from 3000 detached flame images. The NN models effectively learned the complex correlations between these datasets, capturing the underlying physical relationships to enhance prediction accuracy. The study revealed that most velocity information could be derived from OH-PLIF signals, as demonstrated in Fig. 22.

The multi-branch fusion convolutional neural network (MBFCNN) model has achieved remarkable accuracy in reconstructing flow fields from wall pressure data, as illustrated in Fig. 23. Furthermore, by integrating a super-resolution reconstruction model that combines bicubic interpolation, densely connected convolutional neural networks, and generative adversarial networks, the model effectively eliminates image smoothing artefacts. This enhancement results in flow field images with significantly richer and more detailed features.

4.2 Flame reconstruction

Combustion flames exhibit complex, dynamic behaviors that are highly influenced by factors such as fuel type, combustion chamber geometry, and flow conditions. Flame reconstruction could facilitate understanding the fundamental mechanisms of combustion.

By using pressure data and hydrogen injection pressure data, the reparameterized convolutional neural network (RepCNN) model showed a 39.9% improvement in inference speed, with a speed of 2.83 ms per frame achieved, as presented in Fig. 24 [107]. Inspired by the RepVGG network, traditional convolution and deconvolution operations are replaced with reparameterized convolution and reparameterized deconvolution, thereby enhancing the accuracy.

Huang et al. [108] proposed a real-time prediction method for 3D flame evolution and employed the time-resolved volumetric tomography (VT) technique to generate high-quality 3D flame training data. A hybrid CNN-LSTM model was designed to predict flame structure evolution based on historical 2D projections of the flame, as illustrated in Fig. 25. This model

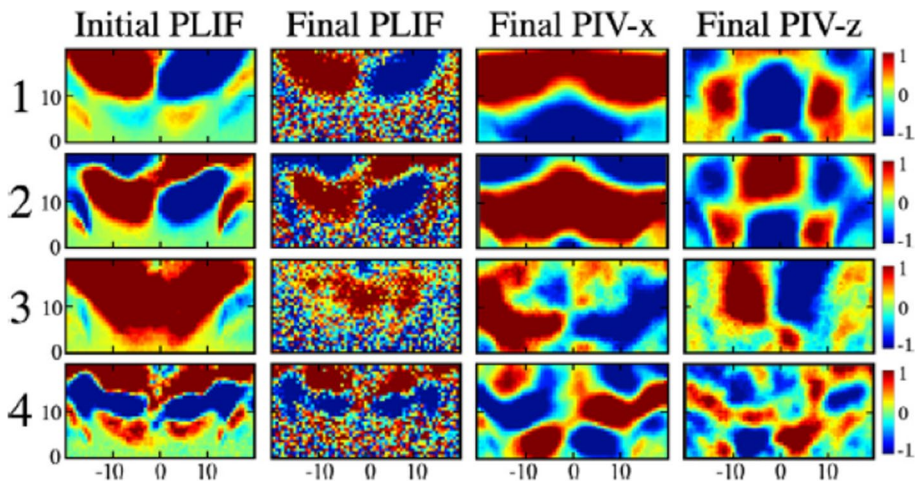


Fig. 22 The four highest-energy ANN weight vectors. Rows indicate weight index [106]

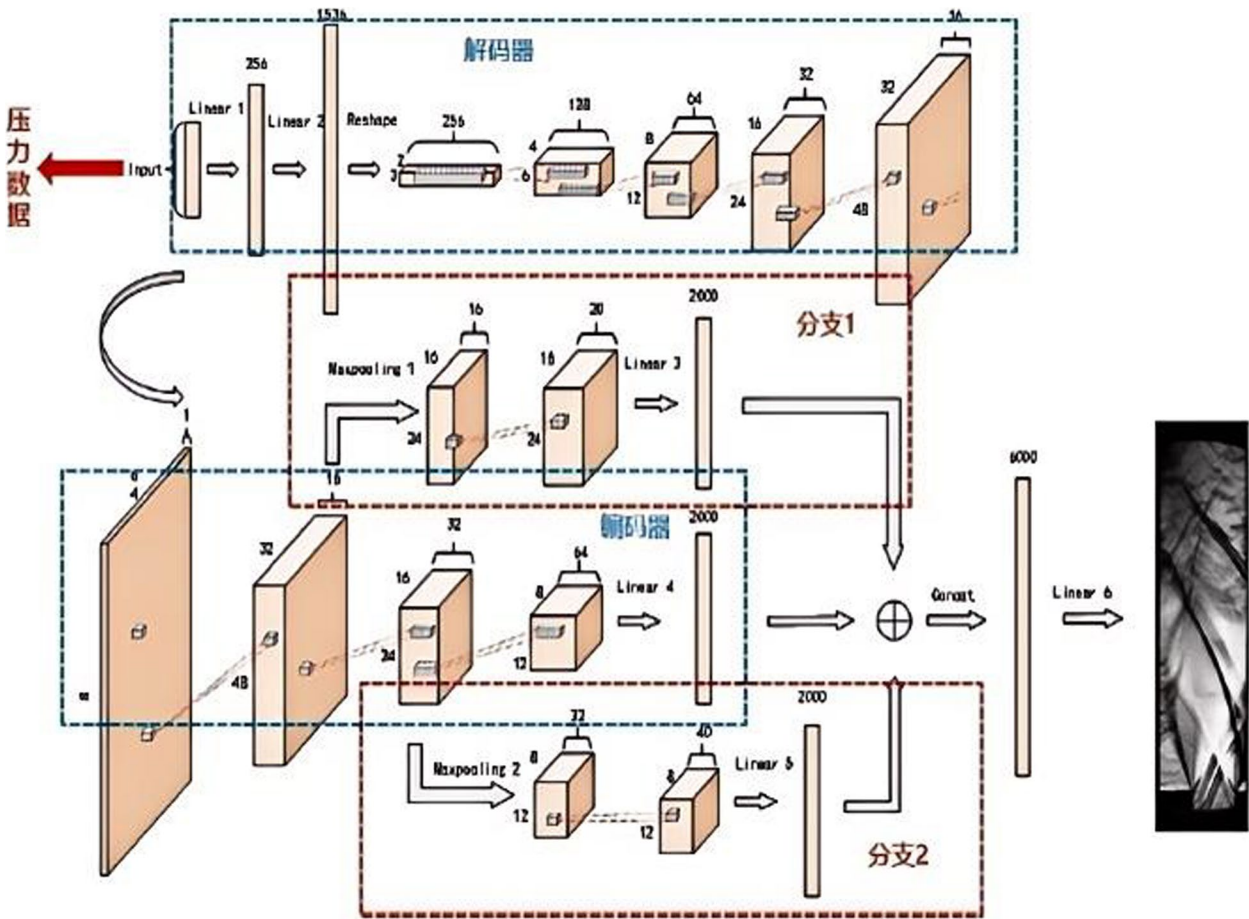


Fig. 23 Structure of MBFCNN model [107]

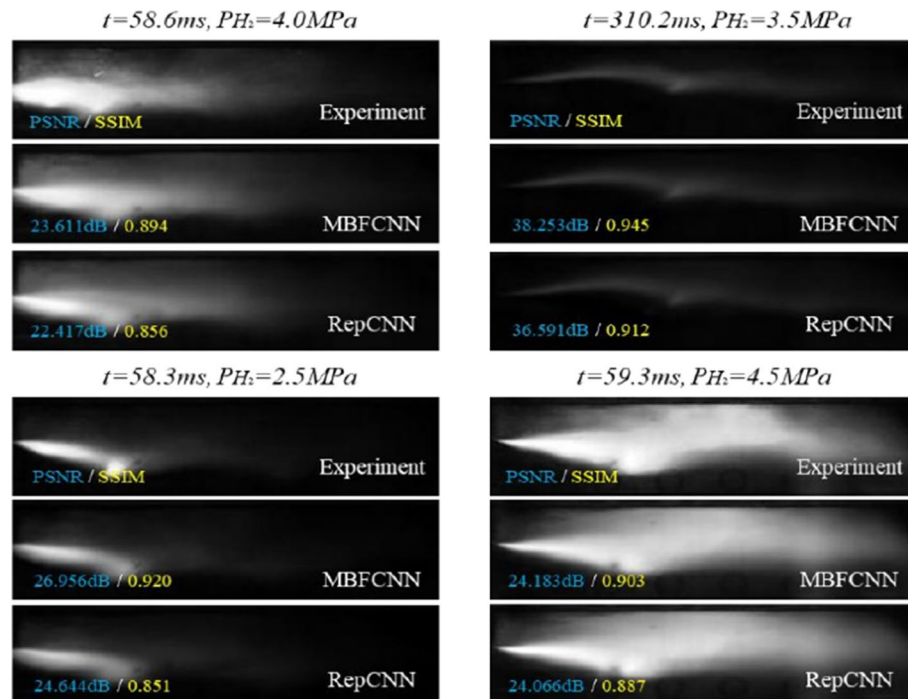


Fig. 24 Initial and final reconstruction flames [107]

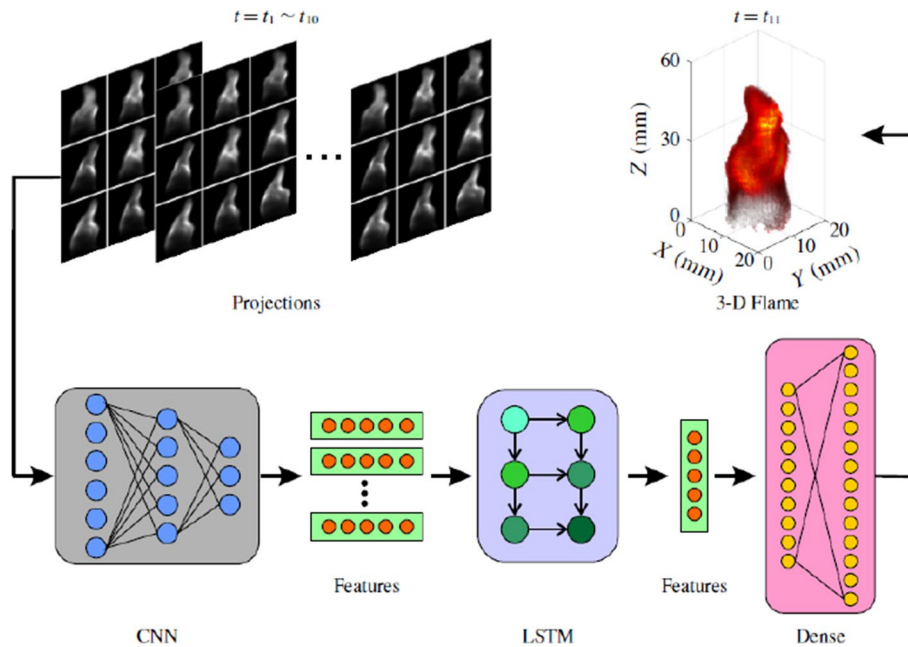


Fig. 25 Schematic of the CNN-LSTM model [108]

could accurately predict the evolution of laminar diffusion flames and typical non-premixed swirl-stabilized flames on a millisecond time scale.

4.3 Combustion state prediction

Combustion processes are inherently complex, involving dynamic interactions between fuel, oxidizers, and

operating conditions. Predicting key combustion parameters, such as flame temperature, oxygen content, pollutant emissions, and pressure fluctuations, enables better control over the process.

Recent advancements in ML techniques have revolutionized the field of combustion state prediction. For instance, Liu and Chen [109] developed an intelligent hydrogen flame detection technology based on ANNs. Predictions were further improved by utilizing a multi-layer deep belief network (DBN), which better captured nonlinear features in combustion processes, such as oxygen content variations during combustion [110].

Other studies, such as Rubiales et al. [111] and Matthaoui et al. [112], showcase the application of ML techniques, like NNs, PCA, and GP, to monitor combustion conditions, detect thermocouple failures, and model complex combustion dynamics, as demonstrated in Fig. 26. In Ref. [95], ensemble ML techniques were applied to diagnose combustion in gas turbines. By combining multiple models, the method enhances prediction accuracy through consensus, reducing errors. In Ref. [112], Gaussian processes were used to model combustion dynamics. This probabilistic approach captures complex relationships and uncertainties, leading to precise predictions while inherently handling outliers and noise, ensuring robustness.

He et al. [113] employed VGG19 for flame state classification and a vid2vid model to synthesize perturbed flame images for robustness evaluation. By quantifying the degree of flame stabilization, the classification results were further refined, significantly reducing the misjudgment of flame conditions. As a result, the method achieved a classification accuracy of 95.44% and a prediction accuracy of over 84% within 2 ms prior to flameout.

Yan [114] proposed a deep semi-supervised anomaly detection (deepSSAD) method, combining DL for feature

extraction and one-class classification with strong performance. Through unsupervised feature extraction, the model could filter out noise and irrelevant data points, improving data quality (Fig. 27).

4.4 Combustion stability prediction

Combustion instability can lead to various operational issues, such as increased emissions, reduced efficiency, and even catastrophic failures, making its early detection and prediction critical in preventing damage to equipment and minimizing operational downtime. Accurately predicting combustion stability [115, 116] enables operators to take proactive measures to maintain stable combustion, thereby improving overall performance and extending the lifespan of combustion equipment.

NN methods effectively extracted and replicated flame features associated with combustion oscillations, enabling a clear distinction between combustion oscillations and stable conditions. Notably, the application of transfer learning with approximately 35% of a new dataset further enhanced the prediction performance for time-averaged images, surpassing that achieved with instantaneous images [117].

A stacked sparse autoencoder-based deep neural network is utilized to monitor combustion stability through flame imaging [118], as shown in Fig. 28. The sparse autoencoder effectively extracts meaningful features from flame images, capturing subtle changes. This feature extraction process inherently filters out noise and irrelevant data, enhancing the model's robustness. The DNN then leverages these refined features to improve prediction accuracy, effectively identifying combustion instabilities with high precision and reducing false alarms.

Supervised learning was effectively utilized to classify combustion conditions in a Honeywell GTCP85 gas turbine, based on the influence of fuel ratios on engine

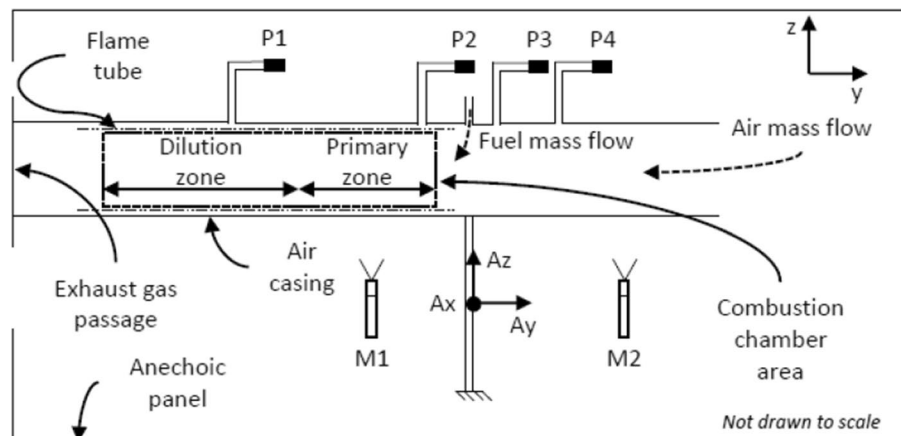


Fig. 26 Schematic diagram of the combustion test rig [112]

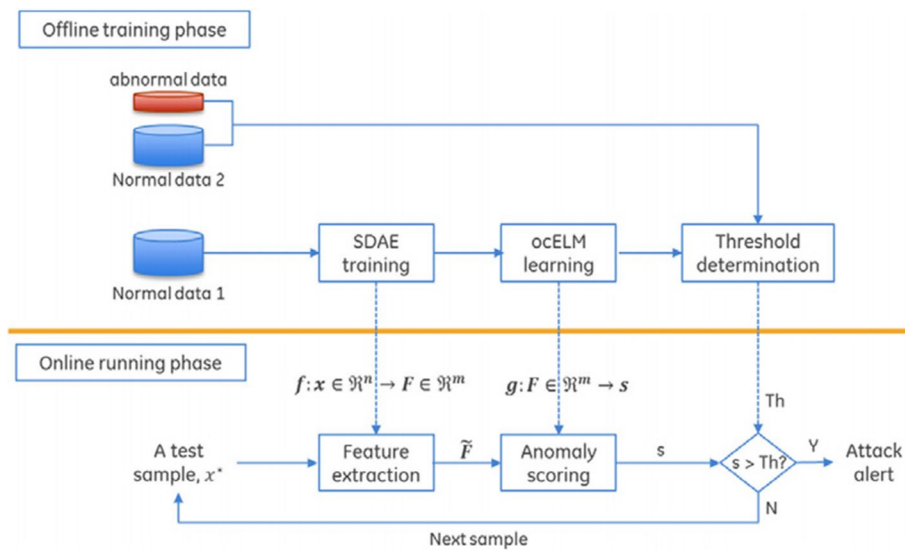


Fig. 27 The overall workflow of the approach [114]

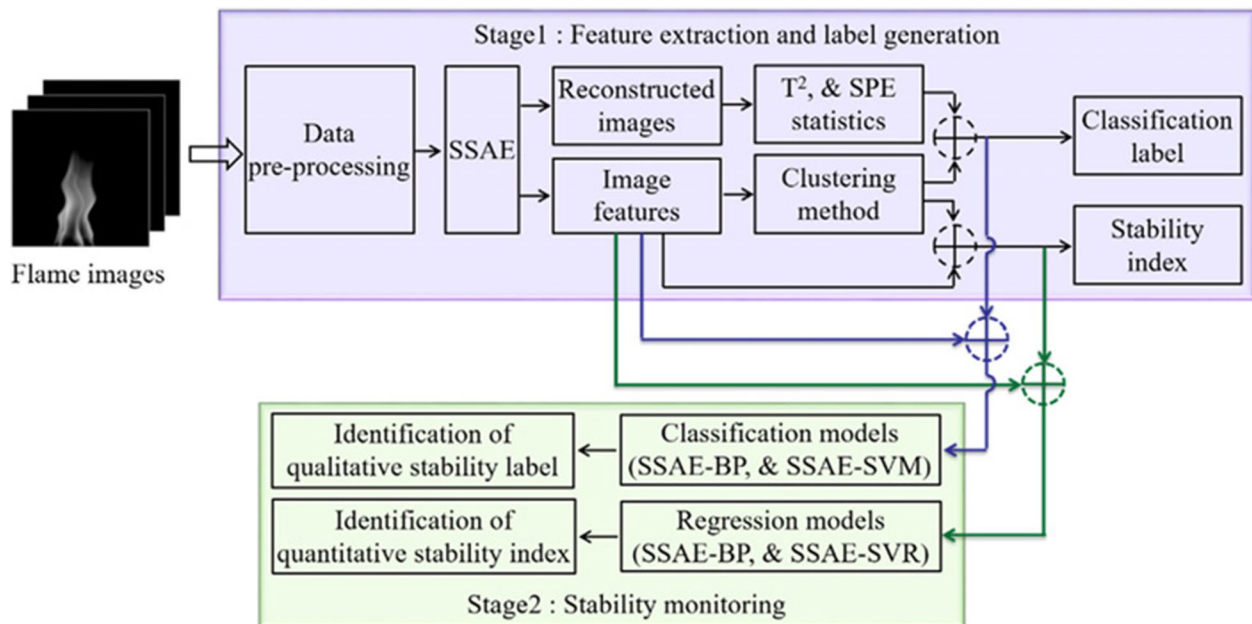


Fig. 28 The strategy of the stacked sparse autoencoder for combustion stability monitoring [118]

performance [119]. Akintayo et al. [120] developed a deep convolutional selective autoencoder for early detection of combustion instability in high-speed flame videos, as presented in Fig. 29. The model selectively masked stable flames while allowing unstable frames to pass, enabling the detection of subtle instability signs during transitions, as demonstrated in Fig. 30.

Hidden Markov models (HMM) were applied to enhance online prediction of thermoacoustic combustion

instability [121]. Additionally, a Bayesian neural network (BNN) was employed to mitigate uncertainty in predicting thermoacoustic instability. BNN-based predictions achieved a 45% reduction in error margin compared to predictions relying solely on operating conditions [122], as shown in Fig. 31.

The CRNN analyses time-series data from a limited number of sensors to detect thermoacoustic instability and predict proximity to instability [123]. Convolutional

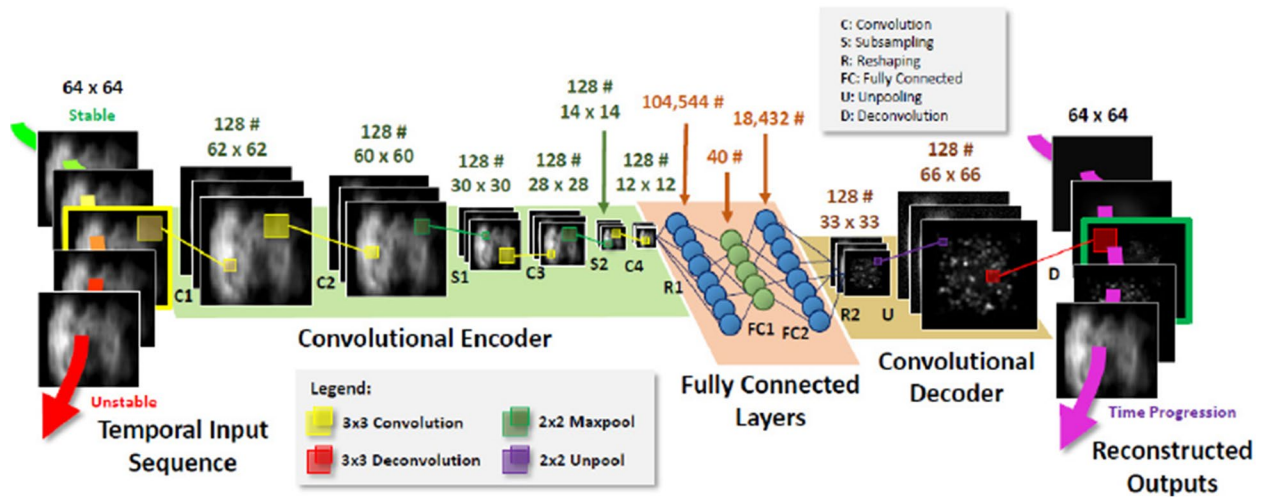


Fig. 29 Structure of the convolutional autoencoder [120]

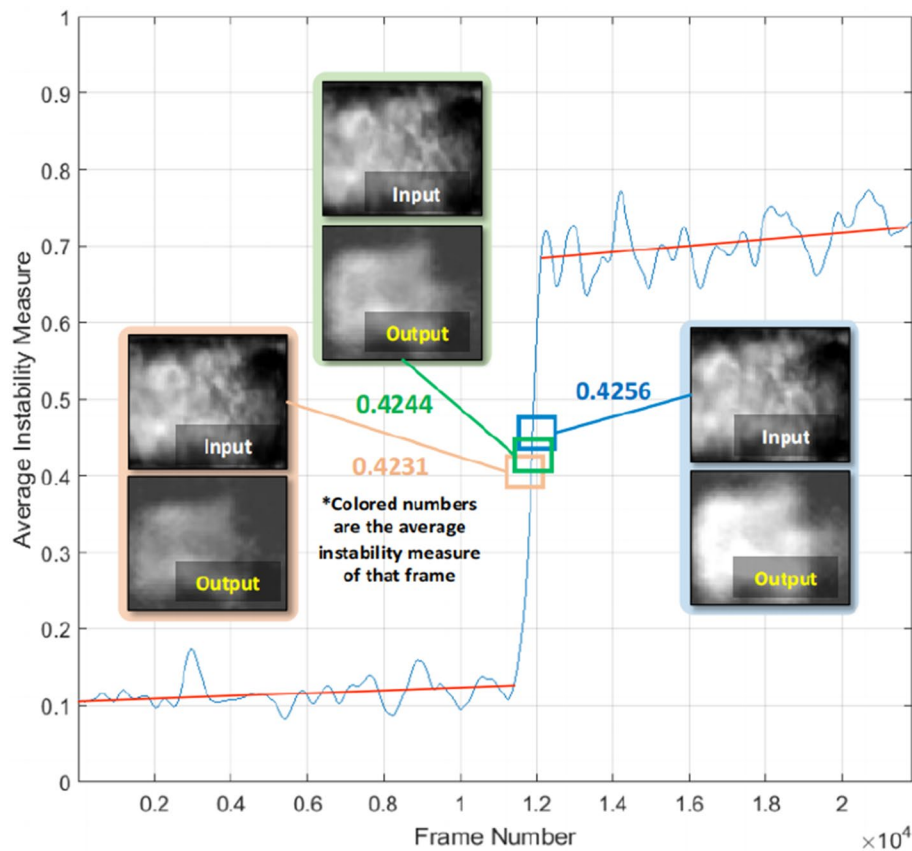


Fig. 30 Adjacency labelling result of transition protocols [120]

layers extract spatial features, while recurrent layers capture temporal dynamics, effectively identifying early signs of instability and enhancing prediction accuracy. Additionally, the model architecture inherently filters out

noise through feature extraction, thereby improving data quality (Fig. 32).

A combination of 3D CNN and LSTM was utilized to monitor combustion instability [124, 125]. Using

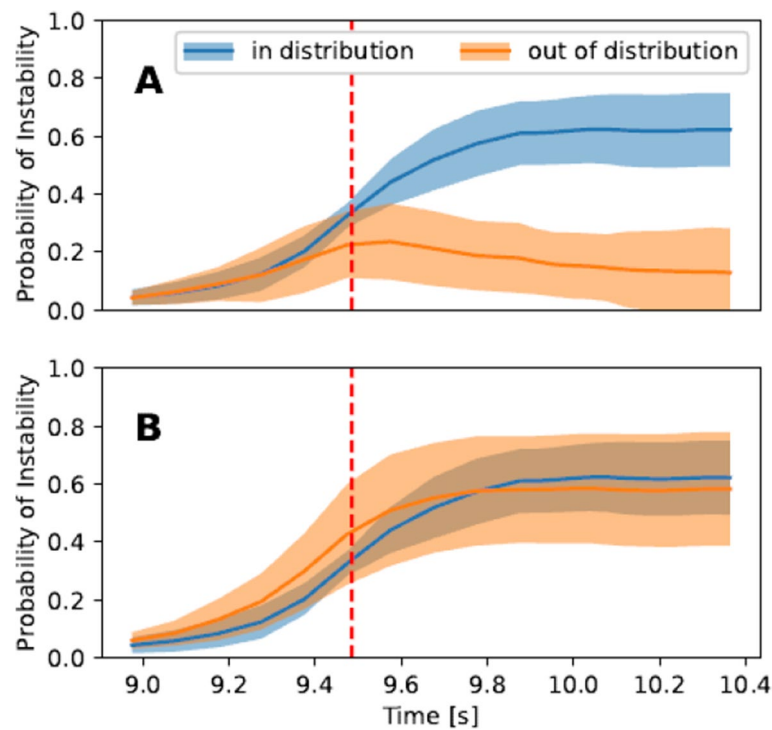


Fig. 31 The predicted probability of an FPCS encountering instability [122]

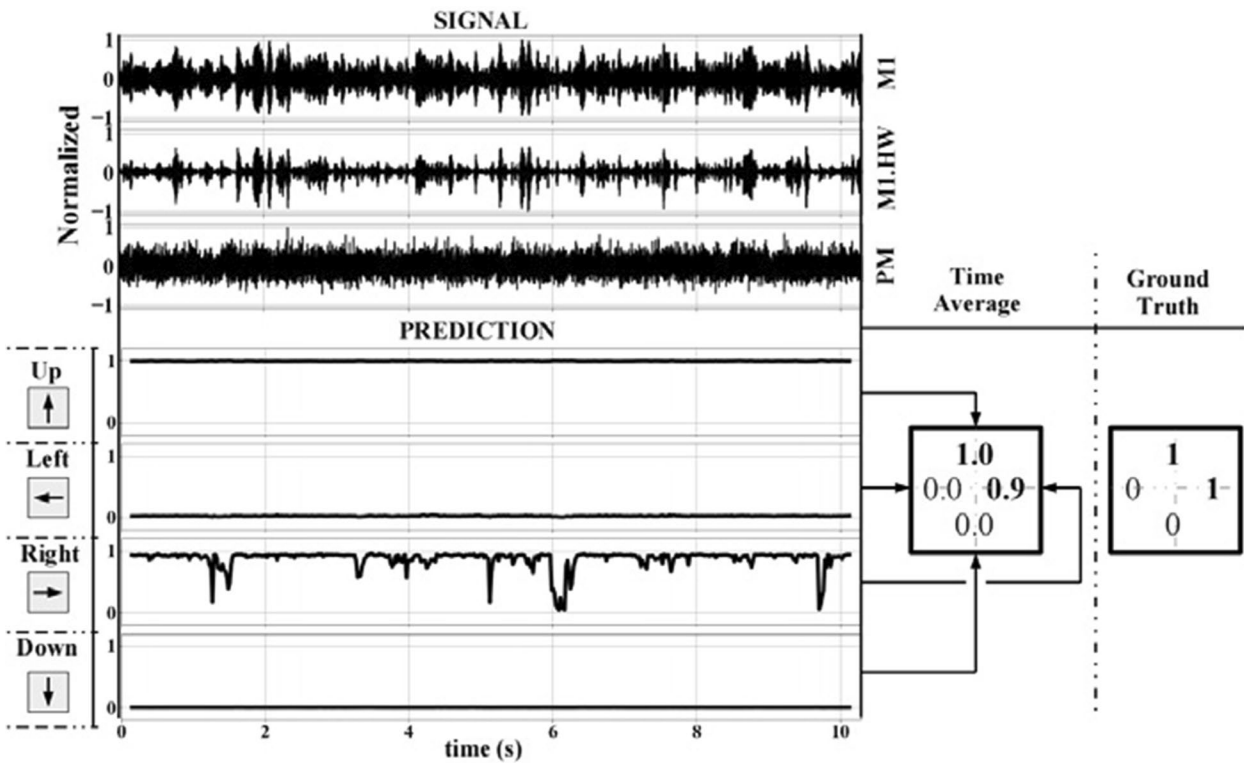


Fig. 32 Rolling-window evaluation of the best-trained network on PCI [123]

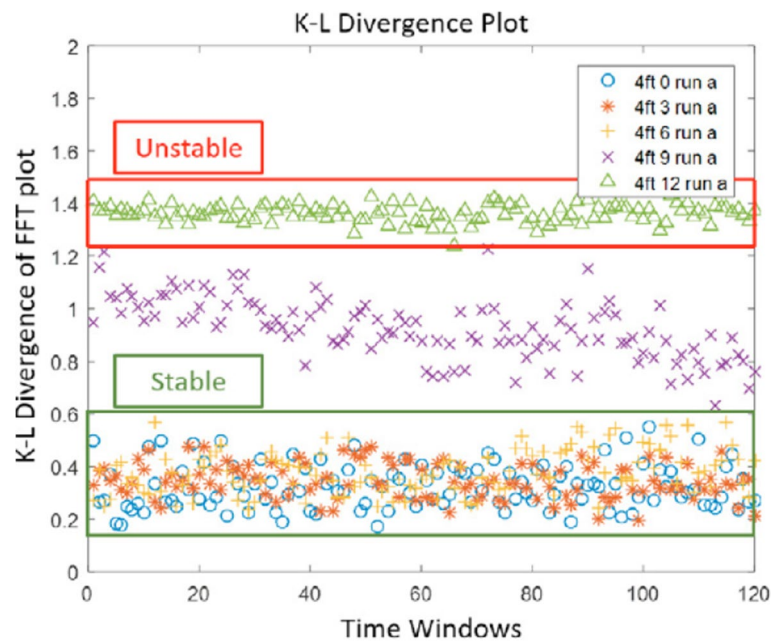


Fig. 33 K-L divergence of FFT plot [124, 125]

multimodal datasets from laboratory-scale combustion systems, Kullback–Leibler divergence was applied for time-series data labelling. By integrating attention mechanisms, the model identified critical time segments that significantly contribute to prediction outcomes (Fig. 33).

Additionally, a 3D convolutional selective autoencoder (3D-CSAE) could be utilized for early detection of combustion instability [126]. This model is capable of learning complex visual and dynamic features. It masks stable flame features while accurately reconstructing unstable flame videos, thereby enhancing the detection of instability (Fig. 34).

PSO-Elman NN model could effectively predict potential instability and associated pollutant emissions (NO_x , CO), demonstrating dual capability in stability monitoring and emission quantification [127].

4.5 Emission prediction

Accurate prediction of emissions (NO_x , CO) is critical for environmental compliance, process optimization, and emission control. Zhang et al. [128] proposed a hybrid multi-output least-squares support vector regression (MLS-SVR)-NN model for multi-emission prediction in biodiesel-hydrogen fueled engines, demonstrating superior accuracy, while Li et al. [129] demonstrated ANN's effectiveness for multi-parameter combustion monitoring (NO_x , dynamics, vibration, and temperature) (Fig. 35).

Potts [130] evaluated ML-based emission prediction performance by comparing self-attention and

inter-sample attention transformer (SAINT) and extreme gradient boosting (XGBoost) against conventional chemistry-based models, demonstrating superior accuracy with ML methods. Wang et al. [131] optimized NN architecture for emission prediction, identifying 24 as the optimal network width ($R^2 > 0.986$), which maximized both prediction accuracy and generalization capability.

4.6 Autoignition prediction

Autoignition poses significant risks in lean-premixed combustion systems, potentially causing flashback and hardware damage during fuel–air mixing. To address this operational challenge, Li et al. [132] developed a hybrid ML model integrating LGR and RNNs for autoignition prediction. The model, validated through experimental fuel–air mixing simulations, successfully identified critical autoignition drivers while achieving robust predictive accuracy.

Table 3 highlights the broad applications of ML in effectively predicting combustion performance under experimental conditions. Algorithms like DNN, CNN, LSTM, and SVM have demonstrated high accuracy in predicting combustion instability, flow field restoration, and emission levels, particularly in handling large datasets and complex, nonlinear relationships.

However, these models often require extensive data, carry a risk of overfitting, and suffer from limited interpretability. These limitations require careful model selection and optimization in practical applications.

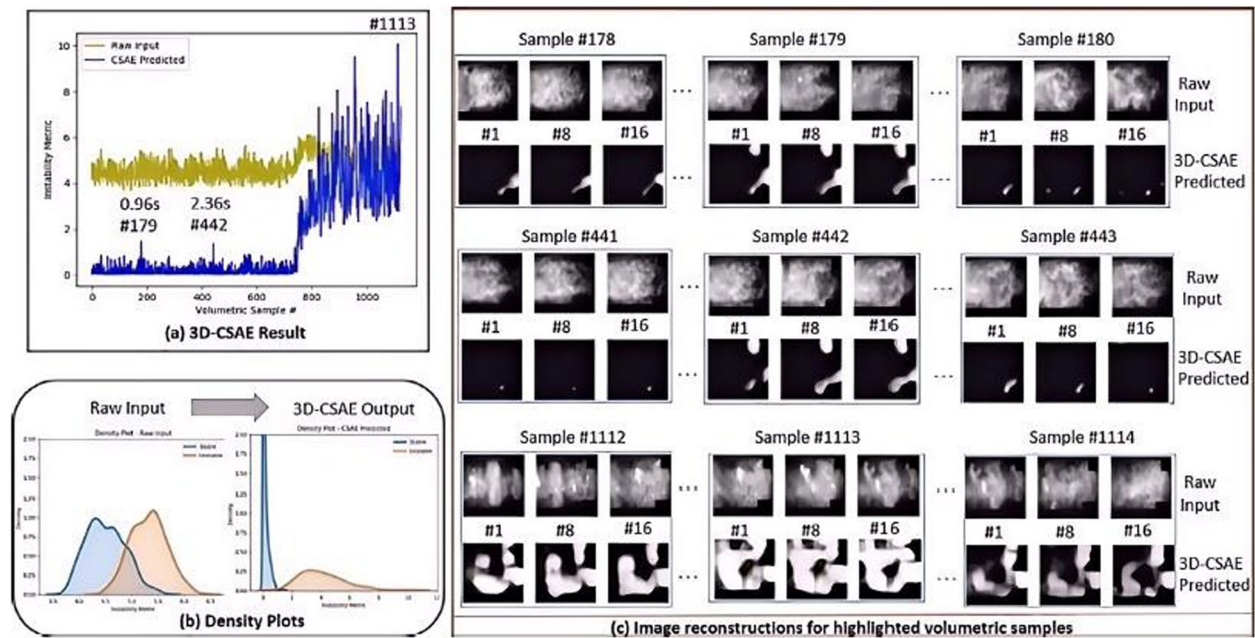


Fig. 34 Results of the 3D-CSAE model [126]

5 Conclusions

The application of ML algorithms in the field of gas turbine combustion has shown significant potential as technology continues to advance. ML algorithms enhance the accuracy of combustion process predictions, especially in capturing complex nonlinear behaviours that traditional physics-based models often miss. This leads to improved simulation efficiency and more accurate performance predictions. Additionally, ML algorithms play a crucial role in optimizing combustion efficiency, reducing harmful emissions, and enhancing engine performance.

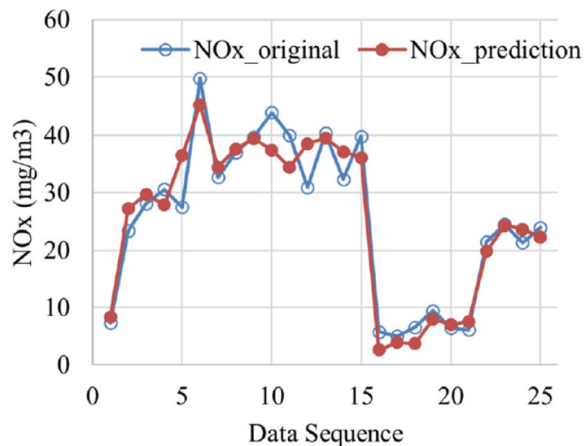


Fig. 35 Model predictions of the NO_x emission [129]

However, the application of ML algorithms also faces several challenges and limitations. Firstly, ML models are highly dependent on large amounts of high-quality training data, which is particularly difficult to obtain in the complex and extreme conditions of aviation engine combustion. The cost and difficulty of data collection are significant hurdles. Secondly, overfitting is a common issue, especially when data samples are limited or of low quality. In such cases, ML models may perform well on the training set but fail to generalize effectively to new and unseen operating conditions. Furthermore, ML models are often criticized as “black boxes” due to their lack of interpretability and transparency, which can hinder trust and practical application, particularly in safety-critical domains like combustion stability.

In conclusion, while data-driven models hold immense promise for revolutionizing combustion science, their practical implementation faces significant hurdles. A primary challenge is the acquisition of high-fidelity experimental data under extreme combustion conditions, such as supersonic flames or sooting pyrolysis. In these environments, harsh temperatures and obscuration often compromise optical diagnostics, making it difficult to obtain reliable and comprehensive data. Additionally, reconciling purely data-driven frameworks with fundamental physical principles remains a critical issue. Models that fail to account for mass and energy conservation or chemical kinetics may violate thermodynamic constraints and lack interpretability, thereby undermining their credibility and applicability.

Table 3 Experimental analysis application

Author	Fuel	Data acquisition methods	Data volume	Method	Accuracy	Experiment
Han et al. [34]	Heavy oil	High-speed camera	6000 sheets	DNN	F1-score = 0.99; <i>R</i> -squared of 0.98	Prediction of combustion instability
Qin et al. [101]	Methane	PIV	600 sets	U-Net CNN	Repair rate of over 90%	Flow field image restoration
Huang et al. [108]	Methane	High-speed camera—time-resolved tomography imaging	10000	CNN and LSTM	<i>R</i> ≈1, RSME≈0.1	Online prediction of 3D flame evolution
Matthaiou et al. [112]	Kerosene	Rolls Royce full cycle combustor test data	5120 data points	GP: Bayesian non-parametric	RMSE < 0.03	Combustion dynamics
He et al. [113]	Methane	High-speed camera	26519 sheets	VGG19_BN	95.44%	Prediction of lean burnout
Yan [114]	Natural gas	One-year exhaust temperature test data of 27 combustors	61666	DeepSSAD	AUC = 0.9706 ± 0.0029	Outlier detection
Zhang et al. [117]	Kerosene	High-speed CCD cameras	67200	SPL and NNs	—	Temperature field prediction
Matthaiou et al. [119]	JetA-1 and biofuel, natural gas	Pressure pulsation signal	14 groups	pattern recognition technology	Classification accuracy reaching 93%	Prediction of combustion instability
McCartney et al. [122]	Hydrogen-natural gas	12 circumferential and 2 axial dynamic pressure tests	30 groups	BNN	Error reduction 45%	Prediction of combustion instability
Cellier et al. [123]	Methane-hydrogen	Combustion noise test	187	CRNN	—	Prediction of combustion instability
Gangopadhyay et al. [124, 125]	Kerosene	High-speed camera and pressure pulsation testing	600000 sheets	3D-CNNs and LSTM	Accuracy approaching 100%	Unstable control and prediction
Potts et al. [130]	Natural gas	SGT400 gas turbine test data	37204 groups	XGBoost	RMSE: NO _x < 4.48 ± 3.65	Emission prediction
Wang et al. [131]	Kerosene	Gas analyzer	210 groups	NN	Correction coefficient of determination > 0.986,	Emission prediction
Li et al. [132]	Acetylene diluted with N ₂	Test data	465 pieces	LGR and RNNs	Prediction accuracy of 88%	Spontaneous combustion prediction

To surmount these challenges and advance the field, future research should prioritize the development of hybrid methodologies that integrate data-driven approaches with established physical principles. For instance, embedding governing equations as neural network constraints, utilizing physics-informed loss functions, or constructing multi-fidelity architectures that combine sparse experimental data with low-cost simulations can significantly enhance model performance. These integrated approaches not only improve model generalizability and compliance with physical laws but also help mitigate data scarcity issues, offering a robust and reliable pathway toward predictive, trustworthy combustion modeling that can drive innovation and progress in combustion science and engineering.

Authors' contributions

Da Mo was responsible for literature review and the initial drafting of the manuscript. Yuzhen Lin oversaw the overall direction and contributed to the revision and refinement of the paper. Yixiong Liu conducted literature collection and participated in revising the initial draft. Yuchen Wang handled the proofreading of the entire manuscript and revised Chapter 2. Ziyu Qin focused on the revision and supplementation of the PINN-related sections. Xiao Han, as the corresponding author, conducted the final proofreading and review of the manuscript.

Funding

The authors would like to thank AECC Shenyang Engine Research Institute and Beihang University for the funding and support. This research was also funded by the National Natural Science Foundation of China (52476099), Science Center for Gas Turbine Project (P2022-A-II-006-001), the Key Laboratory of Pre-Research Management Centre(2023-JCJQ-LB-063-0306) and the Fundamental Research Funds for the Central Universities (501XTCX2023146001).

Declarations

Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Received: 4 December 2024 Revised: 1 August 2025 Accepted: 26 August 2025

Published online: 12 December 2025

References

- Ihme M, Chung WT, Mishra AA (2022) Combustion machine learning: principles, progress and prospects. *Prog Energy Combust Sci* 91:101010
- Ihme M, Chung WT (2024) Artificial intelligence as a catalyst for combustion science and engineering. *Proc Combust Inst* 40:105730
- Liang XM, An JM, Cao XH et al (2023) Classification of combustion state of sintering flame based on CNN-transformer dual-stream network. *J Appl Optics* 44(5):1030–1036
- Legendre AM (1805) *Nouvelles méthodes pour la détermination des orbites des Comètes*. Courcier, Paris
- Rumelhart DE, Hinton GE, Williams RJ (1986) Learning representations by back-propagating errors. *Nature* 323(6088):533–536
- Pearson K (1901) On lines and planes of closest fit to systems of points in space. *Lond Edinb Dublin Philos Mag J Sci* 2(11):559–572
- Lloyd S (1982) Least squares quantization in PCM. *IEEE Trans Inf Theory* 28(2):129–137
- Watkins CJCH (1989) *Learning from delayed rewards*. Dissertation, King's College
- Arrhenius S (1889) Über die dissociationswärme und den einfluss der temperatur auf den dissociationsgrad der elektrolyte. *Z Phys Chem* 4(1):96–116
- Fröhlich J, von Terzi D (2008) Hybrid LES/RANS methods for the simulation of turbulent flows. *Prog Aerosp Sci* 44(5):349–377
- Spalart PR (2000) Strategies for turbulence modelling and simulations. *Int J Heat Fluid Flow* 21(3):252–263
- Spalart PR, Deck S, Shur ML et al (2006) A new version of detached-eddy simulation, resistant to ambiguous grid densities. *Theor Comput Fluid Dyn* 20(3):181–195
- Li W, He X, Xu Y et al (2022) Modeling and simulation of a gas turbine based on series form intelligent fusion algorithm. *J Chinese Soc Power Eng* 42(6):507–512
- Li TY (2022) *Rapid prediction of combustion dynamic field and optimization of anti-temperature in cold-fired boiler based on CFD and POD*. Dissertation, Southeast University
- Xing J, Luo K, Pitach H et al (2019) Predicting kinetic parameters for coal devolatilization by means of artificial neural networks. *Proc Combust Inst* 37:2943–2950
- Kempf A, Flemming F, Janicka J (2005) Investigation of lengthscales, scalar dissipation, and flame orientation in a piloted diffusion flame by LES. *Proc Combust Inst* 30(1):557–565
- Huang ZJ, Xu T, He HH et al (2024) Modeling and simulation of a gas turbine based on series form intelligent fusion algorithm. *J Chinese Soc Power Eng* 44(4):520–527
- Owoyele O, Kundu P, Ameen MM et al (2020) Application of deep artificial neural networks to multi-dimensional flamelet libraries and spray flames. *Int J Engine Res* 21(1):151–168
- Zhang Y, Xu S, Zhong S et al (2020) Large eddy simulation of spray combustion using flamelet generated manifolds combined with artificial neural networks. *Energy and AI* 2:100021
- Gao Z, Jin T, Song CC et al (2021) Application of artificial neural networks to supercritical flamelet model. *J Zhejiang Univ (Eng Sci)* 55(10):1968–1977
- Zhu T (2020) Complex reaction processes in combustion unraveled by neural network-based molecular dynamics simulation. *Nat Commun* 11:5713. <https://doi.org/10.1038/s41467-020-19497-z>
- Christo FC, Masri AR, Nebot EM (1996) Artificial neural network implementation of chemistry with PDF simulation of H_2/CO_2 flames. *Combust Flame* 106(4):406–427
- Ji WQ, Deng SL (2021) Autonomous discovery of unknown reaction pathways from data by chemical reaction neural network. *J Phys Chem A* 125(4):1082–1092
- Zeng JZ, Cao LQ, Xu MY et al (2020) Complex reaction processes in combustion unraveled by neural network-based molecular dynamics simulation. *Nat Commun* 11(1):5713
- Buchheit K, Owoyele O, Jordan T et al (2019) The stabilized explicit variable-load solver with machine learning acceleration for the rapid solution of stiff chemical kinetics. *arXiv: 1905.09395*
- Chen ZX, Iavarone S, Ghiasi G et al (2021) Application of machine learning for filtered density function closure in MILD combustion. *Combust Flame* 225:160–179
- Liu YX, Cong PH, Wu YW et al (2021) Failure analysis and design optimization of shrouded fan blade. *Eng Fail Anal*, 122:105208
- Guo Y, Liu YX, Wu YW et al (2021) Design optimization and burst speed prediction of a Ti2AlNb blisk. *Int J Aerosp Eng*. 2021 : 3290518. <https://doi.org/10.1155/2021/3290518>
- Liu YX, Nalianda D, Mo D et al (2021) Multi-objective optimization of a three-shaft high bypass ratio engine for EIS2050. In: *Proceedings of the Global power and propulsion society, Global Power and Propulsion Society, Xi'an, 18–20 October 2021*
- Liu YX (2021) *Optimization of hybrid electric propulsion system*. Dissertation, Cranfield University
- Niu PF, Ma YP, Zhang XX et al (2019) Research and application on combustion optimization on combustion optimization of coal-fired boiler in thermal power plant based on artificial intelligence technology. *Chinese J Int Sci Tech* 1(2):163–170
- Di Mauro A, Chen H, Sick V (2019) Neural network prediction of cycle-to-cycle power variability in a spark-ignited internal combustion engine. *Proc Combust Inst* 37(4):4937–4944
- Kuzhaglyeva N, Thabet A, Signh E et al (2021) Using deep neural networks to diagnose engine pre-ignition. *Proc Combust Inst* 38(4):5915–5922
- Han ZZ, Li J, Zhang B et al (2021) Prediction of combustion state through a semi-supervised learning model and flame imaging. *Fuel* 289:119745
- Lv Y, Qi X, Zheng X et al (2024) Unsupervised quantitative judgment of furnace combustion state with CBAM-SCAE-based flame feature extraction. *J Energy Inst* 116:101733
- Yu J F, Lian C Y, Tang T, et al (2024) Construction and validation of a wide-domain engine flamelet combustion model based on deep learning table building. *Chinese J Theor Appl Mech* 56(3):723–739
- Chen E D, Song H Y, Guo M M et al (2023) Calculation method of super-sonic hydrogen zero-dimensional ignition based on segmentation-combination residual neural network. *J Propuls Technol* 44(12):91–101
- Zheng J, Wang HO, Chen G et al (2023) Machine learning-based wall model for turbulent boundary layer combustion. *J Eng Thermophys* 44(9):2600–2607
- Lapeyre CJ, Misdariis A, Cazard N et al (2019) Training convolutional neural networks to estimate turbulent sub-grid scale reaction rates. *Combust Flame* 203:255–264
- Nikolaou ZM, Chrysostomou C, Vervisch L et al (2019) Progress variable variance and filtered rate modelling using convolutional neural networks and flamelet methods. *Flow Turbul Combust* 103:485–501
- Fukami K, Fukagata K, Taira K (2019) Super-resolution reconstruction of turbulent flows with machine learning. *J Fluid Mech* 870:106–120
- Klavaris G, Xu M, Hill C et al (2024) Tuning of generalized k-omega turbulence model by using adjoint optimization and machine learning for gas turbine combustor applications. *J Eng Gas Turbines Power* 146(8):081014
- Bode M, Gauding M, Lian Z et al (2021) Using physics-informed enhanced super-resolution generative adversarial networks for subfilter modeling in turbulent reactive flows. *Proc Combust Inst* 38:2617–2625
- Karthik D (2021) Perspectives on machine learning-augmented Reynolds-averaged and large eddy simulation models of turbulence. *Phys Rev Fluids* 6(5):050504

45. Shin JS, Xing V, Pfitzner M et al (2023) Probabilistic deep learning of turbulent premixed combustion. *AIP Adv* 13:085110
46. Shin JS, Ge YP, Lampmann A et al (2021) A data-driven subgrid scale model in large eddy simulation of turbulent premixed combustion. *Combust Flame* 231:111486
47. Lapeyre CJ, Misdariis A, Cazarda N et al (2019) Training convolutional neural networks to estimate turbulent sub-grid scale reaction rates. *Combust Flame* 203:255–264
48. Shateri A, Yang ZY, Xie JF (2025) Machine learning-based prediction of species mass fraction and flame characteristics in partially premixed turbulent jet flame. *Phys Fluids* 37:075219. <https://doi.org/10.1063/5.0277024>
49. Seltz A, Domingo P, Vervisch L et al (2019) Direct mapping from LES resolved scales to filtered-flame generated manifolds using convolutional neural networks. *Combust Flame* 210:71–82
50. Wan K, Barnaud C, Vervisch L et al (2020) Machine learning for detailed chemistry reduction in DNS of a syngas turbulent oxy-flame with side-wall effects. *Proc Combust Inst* 38(2):2825–2833
51. Dalakoti DK, Wehrfritz A, Savard B et al (2021) An a priori evaluation of a principal component and artificial neural network based combustion model in Diesel engine conditions. *Proc Combust Inst* 38(2):2701–2709
52. Lu Z, Metghalchi H (2023) Prediction of laminar burning speed of propane/hydrogen/air mixtures using power-law correlation and two machine learning models. *ASME Open J Eng* 2:021038
53. Molina S, Novella R, Gomez-Soriano J et al (2021) New combustion modelling approach for methane-hydrogen fueled engines using machine learning and engine virtualization. *Energies* 14(21):6732
54. Banta N, Patrick N, Offole F et al (2024) Machine learning models for the prediction of turbulent combustion speed for hydrogen-natural gas spark ignition engines. *Heliyon* 10(9):e30497
55. Malsagov MY, Mikhachenko EV, Karandashev IM et al (2024) Machine learning methods for modeling the kinetics of combustion in problems of space safety. *Acta Astronaut* 225:656–663
56. Li N, Girhe S, Zhang M et al (2024) A machine learning method to predict rate constants for various reactions in combustion kinetic models. *Combust Flame* 263:113375
57. Cao S, Zhang H, Liu H et al (2024) Optimization of kinetic mechanism for hydrogen combustion based on machine learning. *Front Chem Sci Eng* 18(11):136
58. Sadeq AM, Moghaddam AH, Sleiti AK et al (2024) Development of machine learning models for studying the premixed turbulent combustion of gas-to-liquids (GTL) fuel blends. *Korean J Chem Eng* 41(2):479–494
59. Ihme M, Schmitt C, Pitsch H (2009) Optimal artificial neural networks and tabulation methods for chemistry representation in LES of a bluff-body swirl-stabilized flame. *Proc Combust Inst* 32(1):1527–1535
60. Li S, Yang B, Qi F (2016) Accelerate global sensitivity analysis using artificial neural network algorithm: case studies for combustion kinetic model. *Combust Flame* 168:53–64
61. Lu JH, Zhang HM, Yu JH et al (2021) Predicting rate constants of hydroxyl radical reactions with alkanes using machine learning. *J Chem Inf Model* 61:4259–4265
62. Ji W, Deng S (2021) KiNet: A deep neural network representation of chemical kinetics. *arXiv:2108.00455*
63. Liu M, Grinberg Dana A, Johnson MS et al (2021) Reaction mechanism generator v3.0: advances in automatic mechanism generation. *J Chem Inf Model* 61(6):2686–2696
64. Castellanos L, SM Freitas R, Parente A et al (2023) Deep learning dynamical latencies for the analysis and reduction of combustion chemistry kinetics. *Phys Fluids* 35(10):107143
65. Han X, Jia M, Chang Y, Li Y (2022) An improved approach towards more robust deep learning models for chemical kinetics. *Combust Flame* 238:111934
66. Haghshenas M, Mitra P, Santo N et al (2021) Acceleration of chemical kinetics computation with the learned intelligent tabulation (LIT) method. *Energies* 14(23):7851
67. Pulga L, Bianchi GM, Falfari S et al (2020) A machine learning methodology for improving the accuracy of laminar flame simulations with reduced chemical kinetics mechanisms. *Combust Flame* 216:72–81
68. Staszak M (2023) Artificial intelligence in the modeling of chemical reactions kinetics. *Phys Sci Rev* 8(1):51–72
69. Yao S, Wang B, Kronenburg A et al (2021) Conditional scalar dissipation rate modeling for turbulent spray flames using artificial neural networks. *Proc Combust Inst* 38:3371–3378
70. Yao S, Wang B, Kronenburg A et al (2020) Modeling of sub-grid conditional mixing statistics in turbulent sprays using machine learning methods. *Phys Fluids* 32:115124. <https://doi.org/10.1063/5.0027524>
71. Henry de Frahan MT, Yellapantula S, King R et al (2019) Deep learning for presumed probability density function models. *Combust Flame* 208:436–450
72. Lapeyre CJ, Misdariis A, Cazard N et al (2019) Training convolutional neural networks to estimate turbulent sub-grid scale reaction rates. *Combust Flame* 203:255–264
73. Nikolaou ZM, Chrysostomou C, Minamoto Y et al (2021) Evaluation of a neural network-based closure for the unresolved stresses in turbulent premixed V-flames. *Flow Turbul Combust* 106:331–356
74. Yellapantula S, Perry BA, Grout RW (2021) Deep learning-based model for progress variable dissipation rate in turbulent premixed flames. *Proc Combust Inst* 38(2):2929–2938
75. Guo M, Chen H, Tian Y et al (2023) Flame reconstruction of hydrogen fueled-scrumjet combustor based on multisource information fusion. *Int J Hydrogen Energy* 48(80):31350–31365. <https://doi.org/10.1016/j.ijhydene.2023.04.311>
76. Song T, Liu H D, Yue H et al (2025) Identification of combustion modes in dual-mode combustors across configurations using domain adversarial neural network. *J Propuls Technol* 46(2):2312004
77. Ricci F, Avana M, Mariani F (2024) A deep learning method for the prediction of pollutant emissions from internal combustion engines. *Appl Sci* 14(21):9707
78. Pachauri N (2024) An emission predictive system for CO and NO_x from gas turbine based on ensemble machine learning approach. *Fuel* 366:131421
79. Chen L, Zhang Q, Zhu M et al (2024) A convolutional neural network prediction model for aviation nitrogen oxides emissions throughout all flight phases. *Sci Total Environ* 929:172432
80. Chen Z, Pu Y (2024) Prediction of emissions from gas turbine power generation on GWO-XGBoost-Sobol. *J Mech Sci Technol* 38(3):1547–1556
81. Chung WT, Mishra AA, Perakis N et al (2021) Data-assisted combustion simulations with dynamic submodel assignment using random forests. *Combust Flame* 227:172–185
82. Nassini PC, Pampaloni D, Meloni R et al (2021) Lean blow-out prediction in an industrial gas turbine combustor through a LES-based CFD analysis. *Combust Flame* 229:111391
83. Ahmed E, Yong H (2018) Prediction of lean blowout performance of gas turbine combustor based on flow structures. *Aeronaut J* 122(1248):238–259
84. Bahashwan AA, Ibrahim R, Omar M et al (2024) Supervised learning-based multi-site lean blowout prediction for dry low emission gas turbine. *Expert Syst Appl* 244:123035
85. Raju Hasti V, Navarkar A et al (2021) A data-driven approach using machine learning for early detection of the lean blowout. *Energy and AI* 5:100099
86. Liang XM, An JM, Cao XH et al (2023) Classification of combustion state of sintering flame based on CNN-transformer dual-stream network. *J Appl Optics* 44(5):1030–1036
87. Zhang TH, Yi YX, Xu YF et al (2022) A multi-scale sampling method for accurate and robust deep neural networks to predict combustion chemical kinetics. *Combust Flame* 245:112319
88. Schweidtmann AM, Rittig JG, König A et al (2020) Graph neural networks for prediction of fuel ignition quality. *Energy Fuels* 34(9):11395–11407
89. Choi J, Hong S, Park N et al (2023) Gread: Graph neural reaction-diffusion networks. *arXiv: 2211.14208*
90. Karniadakis GE, Kevrekidis IG, Lu L et al (2021) Physics-informed machine learning. *Nat Rev Phys* 3:422–441
91. Bradley W, Gusmão GS, Medford AJ et al (2021) Training stiff dynamic process models via neural differential equations. *Computer Aided Chem Eng* 49:1741–1746
92. Ji W, Qiu W, Shi Z et al (2021) Stiff-PINN: physics-informed neural network for stiff chemical kinetics. *J Phys Chem A* 125(18):8098–8106

93. De Florio M, Schiassi E, Furfaro R (2022) Physics-informed neural networks and functional interpolation for stiff chemical kinetics. *Chaos* 32(6):063107
94. Weng YT, Zhou DZ (2022) Multiscale physics-informed neural networks for stiff chemical kinetics. *J Phys Chem A* 126:8534–8543
95. Ngo SI, Lim YI (2021) Solution and parameter identification of a fixed-bed reactor model for catalytic CO₂ methanation using physics-informed neural networks. *Catalysts* 11:1304
96. Aquilanti V, Mundim KC, Elango M et al (2010) Temperature dependence of chemical and biophysical rate processes: phenomenological approach to deviations from Arrhenius law. *Chem Phys Lett* 498:209–213
97. Almeldein A, Van Dam N (2023) Accelerating chemical kinetics calculations with physics informed neural networks. No. ICEF2022-90371
98. Zhang S, Zhang C, Wang B (2024) CRK-PINN: a physics-informed neural network for solving combustion reaction kinetics ordinary differential equations. *Combust Flame* 269:113647
99. Raissi M, Perdikaris P, Karniadakis GE (2019) Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *J Comput Phys* 378:686–707
100. Raissi M (2017) Physics informed deep learning (part I): data-driven solutions of nonlinear partial differential equations. *arXiv:1711.10561*
101. Qin ZY, Zheng DS, Zhou Y C et al (2022) Inpainting PIV flow fields with deep learning. *J Propuls Technol* 43(9):330–337. <https://kns.cnki.net/kcms/detail/11.1813.v.20220310.1532.006.html>
102. Grant I, Pan X (1997) The use of neural techniques in PIV and PTV. *Meas Sci Technol* 8(12):1399–1405
103. Lagemann C, Lagemann K, Schroder W et al (2019) Deep artificial neural network architectures in PIV applications. In: International symposium on particle image velocimetry, Munich, 22–24 July 2019
104. Zhu H R, Gao Q, Wang H P et al (2021) 3D particle reconstruction technology based on machine learning methods. *Experi Fluid Mech* 35(3):88–93. <https://kns.cnki.net/kcms/detail/11.5266.v.20210219.1311.002.html>
105. Gao ZY, Li XY, Ye HW (2020) Optical distortion correction technology for flow velocity measurements based on convolutional neural networks. *Experi Fluid Mech* 49(10):9–18
106. Barveey S, Raman V, Steinberg AM (2020) Extracting information overlap in simultaneous OH-PLIF and PIV fields with neural networks. *Proc Combust Inst* 38(4):6241–6249
107. Chen H (2023) Research on data driven intelligent reconstruction algorithm of scramjet combustion flow field. Dissertation, Southwest University of Science and Technology
108. Huang J, Liu H, Cai W (2019) Online in situ prediction of 3-D flame evolution from its history 2-D projections via deep learning. *J Fluid Mech* 875:R2
109. Liu MY, Chen W (2023) Research on intelligent detection technology of hydrogen flame based on artificial intelligence BP neural network. *Energy Res Utiliz* 4:17–22
110. Liu Y, Fan Y, Chen JH (2017) Flame images for oxygen content prediction of combustion systems using DBN. *Energy Fuels* 31(8):877–8783
111. Rubiales MM, Munoz A, Sanz-bobi M et al (2024) Application of ensemble machine learning techniques to the diagnosis of the combustion in a gas turbine. *Appl Therm Eng* 249:123447
112. Matthaïou I, Khandelwal B, Antoniadou I (2017) Using Gaussian processes to model combustion dynamics. In: 24th international London congress on sound and vibration, London, 23–27 July 2017
113. He T, Wang S, Wei S et al (2022) Prediction method and experimental research on lean burn blow-off based on deep learning. *J Combust Sci Technol* 28(3):304–312
114. Yan W (2020) Detecting gas turbine combustor anomalies using semi-supervised anomaly detection with deep representation learning. *Cogn Comput* 12:398–411
115. Zhou Y, Zhang C, Han X et al (2021) Monitoring combustion instabilities of stratified swirl flames by feature extractions of time-averaged flame images using deep learning method. *Aerosp Sci Technol* 109:106443
116. Qin ZY, Wang XY, Han X et al (2023) Pre-trained combustion model and transfer learning in thermoacoustic instability. *Phys Fluids* 35(3):037117
117. Zhang SH, Wang XY, Zhang C et al (2023) Feature extraction and combustion instabilities prediction of stratified swirling flame. *J Combust Sci Technol* 29(5):483–490
118. Han Z, Hossain MM, Wang Y et al (2020) Combustion stability monitoring through flame imaging and stacked sparse autoencoder based deep neural network. *Appl Energy* 259:114159
119. Matthaïou I, Khandelwal B, Antoniadou I (2016) Towards a condition monitoring scheme for thermoacoustic instability detection and fuel blend performance classification in gas turbine engines using pattern recognition and advanced machine learning. In: 8th European workshop on structural health monitoring, Bilbao, 5–8 July 2016
120. Akintayo A, Lore KG, Sarkar S (2017) Early detection of combustion instabilities using deep convolutional selective autoencoders on Hi-speed flame video. *arXiv:1603.07839*
121. McCartney M, Indlekofer T, Polifke W (2020) Online detection of combustion instabilities using supervised machine learning, ASME GT2020-14834
122. McCartney M, Sengupta U, Juniper M (2021) Reducing uncertainty in the onset of combustion instabilities using dynamic pressure information and Bayesian neural networks. No. GT2021-60283
123. Cellier A, Lapeyre CJ, Öztarlık G et al (2021) Detection of precursors of combustion instability using convolutional recurrent neural networks. *Combust Flame* 233:111558
124. Gangopadhyay T, Tan SY, Locruato A et al (2020) Interpretable deep learning for monitoring combustion instability. *IFAC-PapersOnLine* 53(2):832–837
125. Gangopadhyay T, Locruato A, Michael J B et al (2020) Deep learning algorithms for detecting combustion instabilities. In: Mukhopadhyay A, Sen S, Basu D et al (ed) Proceedings of the dynamics and control of energy systems, Springer, Singapore, 2020
126. Gangopadhyay T, Ramanan V, Akintayo A et al (2021) 3D convolutional selective autoencoder for instability detection in combustion systems. *Energy AI* 4:100067
127. Wang WY (2023) Prediction and analysis of combustion stability of gas turbine based on neural network. *J Chinese Soc Power Eng* 43(7):842–848
128. Zhang L, Zhu G, Chao Y et al (2023) Simultaneous prediction of CO₂, CO, and NO_x emissions of biodiesel-hydrogen blend combustion in compression ignition engines by supervised machine learning tools. *Energy* 282:128972
129. Li S, Zhu H, Zhu M et al (2021) Combustion tuning for a gas turbine power plant using data-driven and machine learning approach. *J Eng Gas Turbines Power* 143(3):031021
130. Potts R, Hackney R, Leontidis G (2023) Tabular machine learning methods for predicting gas turbine emissions. *Mach Learn Knowl Extr* 5:1055–1075
131. Wang Z K, Chen S, Fan W (2023) Effect of neural network width on combustor emission prediction. *Acta Aeronaut Astronaut Sinica* 44(5):126816
132. Li S, Qian W, Liu H et al (2020) Prediction of the autoignition of a fuel jet in a confined turbulent hot coflow using machine learning methods. No. GT2020-15020

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.